

CMS Overall Hospital Quality Star Rating Comprehensive Methodology Report (v5.1)

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Executive Summary

This report presents version 5.1 of the methodology for the Overall Hospital Quality Star Rating, developed by the Yale New Haven Health Services Corporation – Center for Outcomes Research and Evaluation (CORE) under contract to the Centers for Medicare & Medicaid Services (CMS), as implemented in April 2026.[†] This report describes CMS’s approach to constructing a methodology for generating an Overall Hospital Quality Star Rating for each eligible hospital providing acute inpatient and outpatient care and publicly reporting quality information via [Care Compare on Medicare.gov](https://carecompare.cms.gov).

Overview of Project Objective

CMS contracted with CORE to work in collaboration with other contractors to develop a methodology for the Overall Hospital Quality Star Rating. The primary objective of the Overall Hospital Quality Star Rating project is to develop a statistically sound methodology for summarizing information from the existing measures publicly reported via [Care Compare on Medicare.gov](https://carecompare.cms.gov) in a way that is useful and easy to interpret for patients and consumers. [Care Compare on Medicare.gov](https://carecompare.cms.gov) includes information on quality measures for more than 4,000 acute care and critical access hospitals. Consistent with other CMS star rating programs, this methodology assigns each hospital between one and five stars, reflecting the hospital’s overall performance on publicly reported quality measures.

The Overall Hospital Quality Star Rating complements existing efforts, such as the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) star ratings (implemented in April 2015) and will not replace the reporting of any individual quality measures. In what follows, “Overall Star Rating” refers to Overall Hospital Quality Star Rating, unless otherwise noted.

The Overall Star Rating was first introduced and publicly reported via [Care Compare on Medicare.gov](https://carecompare.cms.gov) (then called *Hospital Compare*) in July 2016 and had since been refreshed twelve times, most recently in July 2025. The original Overall Star Rating methodology and subsequent methodology updates were developed and maintained with substantial input from stakeholders and technical experts.

Overall Star Rating methodology version	Releases	Notes
1.0	Confidential dry run: July 2015	Confidential reporting only
2.0	Four releases: July 2016 – October 2017	First published results
3.0	Three releases: December 2017 – January 2020	Minor updates to k-means clustering algorithm
4.1	Five releases: April 2021 – July 2025	Peer grouping incorporated; switch from latent variable model to explicit group score calculation
5.1	1+ release: April 2026 –	Switch from SAS to R for calculation; addition of Safety of Care cap

[†] Version 5.1 of the Overall Star Rating Methodology is the same as Version 5.0. Table 8 of this Methodology Report has been updated since the Version 5.0 upload to provide additional clarity.

Guiding Principles for Developing & Maintaining Overall Star Ratings

Based on a review of the literature, lessons from prior star rating efforts, and the overarching CMS quality strategy, CMS identified the following principles to guide Overall Star Rating work.

1. Be scientifically valid
2. Remain inclusive of hospital and measure information
3. Account for the heterogeneity of available measures and hospital reporting
4. Accommodate changes in the underlying measures
5. Align with Care Compare on Medicare.gov and other CMS programs
6. Ensure transparency of methodological decisions
7. Be responsive to stakeholder input

Since early development of the Star Rating in 2014, CMS has maintained these principles by holding nine multi-stakeholder Technical Expert Panels (TEP), four public input periods, eight National Provider Calls, a confidential dry run, 10 Person & Family Engagement (PFE) Workgroup (WG) meetings, 10 Provider Leadership Work Group (Provider WG) meetings, and participating in a Star Rating summit hosted by the National Quality Forum (NQF). CMS additionally included the Overall Star Rating methodology in the proposed Calendar Year (CY) 2021 Outpatient Prospective Payment System (OPPS)/Ambulatory Surgical Center (ASC) Proposed Rule¹, the CY 2023 OPPS/ASC Proposed Rule², the CY 2025 OPPS/ASC Proposed Rule³, and the CY 2026 OPPS/ASC Proposed Rule⁴ all of which provided additional opportunities for the public to comment on the methodology.

CMS designed several aspects of the Overall Star Rating development process to include the patient and consumer perspective in key methodological and policy decisions. Both the TEP and PFE WG included diverse patient and patient advocate representation. These individuals were generally supportive of CMS's decision to develop a hospital quality star ratings system, expressing its potential value and importance to patients and consumers. CMS engaged a variety of providers including leaders in delivery organizations, such as chief quality officers and chief medical officers, and hospital representatives through our TEP and Provider WG. See [Appendix A](#) for a complete list of TEP and WG representation.

Version 5.1 of the methodology contains one substantive update to the methodology: In alignment with broader efforts within HHS and CMS to underscore the importance of patient safety for both patients and healthcare workers, CMS has implemented a cap limiting hospitals to a maximum rating of 4 stars if they perform poorly on measures of patient safety.

Overview of Methodology

Version 5.1 of the methodology (v5.1) takes a nine-step approach to calculating the Overall Star Rating. In the first step, the measures are selected from [Care Compare on Medicare.gov](#) based on their relevance and importance as determined through stakeholder and expert feedback. Second, the included measure scores are standardized to be consistent in terms of direction and magnitude. Third, these standardized measures are then organized into five groups according to measure type - namely Safety of Care, Mortality, Readmission, Patient Experience, and Timely and Effective Care. Fourth, for each hospital, a simple average of the standardized measure scores is calculated and this score is again standardized to generate a measure group score. In the fifth step, a predetermined weight is applied to each measure group score to calculate a hospital summary score. In the sixth step, the public reporting

threshold is applied, requiring hospitals to have a minimum of three measure groups (one of which must be the Mortality or Safety of Care group) with at least three measures in each of the three groups to receive an Overall Star Rating. In the seventh step, hospitals are grouped into three 'peer groups' based on the number of measure groups for which they report at least three measures. In the eighth step, hospital summary scores are categorized into 5 clusters. In the newly added ninth step, 5 stars are assigned to the highest cluster and 1 star to the lowest cluster; furthermore a 4-star cap is applied to hospitals that score in the lowest quartile on their Safety of Care measure group score and report at least three safety measures.

Conclusion

The overarching goal of the Overall Hospital Quality Star Rating is to improve the usability and interpretability of [Care Compare on Medicare.gov](https://carecompare.cms.gov) for patients and consumers. This report reflects the results of a three-year long effort to assess the Overall Star Rating as implemented and engage stakeholders in enhancements to the methodology to emphasize Safety of Care. The previous methodology (v4.1), released in 2021, was similarly a multi-year effort that included multiple forms of stakeholder feedback. CMS will continue to engage stakeholders and evaluate the Overall Star Rating methodology.

Introduction

Project Objective

CMS contracted with CORE to work in collaboration with other contractors to develop the methodology for the Overall Hospital Quality Star Rating published to [Care Compare on Medicare.gov](https://carecompare.medicare.gov) (formerly known as *Hospital Compare*, and subsequently now Care Compare on Medicare.gov). [Care Compare on Medicare.gov](https://carecompare.medicare.gov) includes information on over 100 quality measures for more than 4,000 Medicare-certified hospitals. The primary objective of the Overall Hospital Quality Star Rating project is to summarize information from the existing quality measures publicly reported via [Care Compare on Medicare.gov](https://carecompare.medicare.gov) in a way that is useful and easy to interpret for patients and consumers through the development of a statistically sound methodology. Consistent with other CMS star rating programs, this methodology assigns each hospital between one and five stars, reflecting the hospital's overall performance on Care Compare quality measures.

The Overall Hospital Quality Star Rating is designed to provide summary information for consumers about existing publicly reported quality data. CMS intends for the Overall Hospital Quality Star Rating to complement existing efforts, such as the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) Star Rating (implemented in April 2015) and will not replace the reporting of any individual quality measures. Throughout the remainder of this report, "Overall Star Rating" refers to the Overall Hospital Quality Star Rating, unless otherwise noted.

This report outlines methodology changes since the Overall Star Rating v4.1, namely the implementation of a 4-star cap for hospitals in the lowest quartile of performance for the Safety of Care measure group reporting at least three Safety of Care measures. Additionally, the Overall Star Rating is now calculated using R instead of SAS.

Why Develop an Overall Star Rating?

In 2014, CMS conducted a review of the literature and prior star rating efforts which supported the notion that patients care about quality information. The results also suggested that patients' use of this information is limited by its complexity and inconsistency in facets of quality that interest them most. Consumers need help understanding hospital quality information and prefer information to be presented in a more condensed and annotated manner to convey the many facets of quality. These key findings are consistent with consumers' priorities of bringing a wide variety of measures together into a single star rating and also point to the need for extensive engagement and education of stakeholders throughout development and implementation.

In addition to patients' and consumers' informational needs, CMS developed the Overall Star Rating methodology to complement the methodologies and goals of other CMS programs and star rating initiatives, including: Dialysis Facility Compare Star Ratings, Home Health Compare Quality of Patient Care Star Ratings, HCAHPS Star Ratings, Nursing Home Compare Star Ratings, Medicare Plan Finder Star Ratings, and Qualified Health Plans (QHPs) Quality Rating System (QRS).

Reevaluation of the Overall Star Rating Methodology

The Overall Star Rating is a summary of certain existing information about hospital quality, which is collected and reported as part of several CMS programs to improve and make transparent the quality of care for hospitals that provide acute inpatient and outpatient care. As the underlying measures reported via [Care Compare on Medicare.gov](https://carecompare.medicare.gov) have been added, updated, and removed, and as stakeholders have begun using the methodology for purposes beyond consumer transparency, including provider quality improvement efforts, refinements to the methodology of the Overall Star Rating are necessary. Since first reporting the Overall Star Rating in July 2016, CMS has maintained an active monitoring and reevaluation process for the methodology, as well as engaged stakeholders for continuous feedback. Based on this ongoing reevaluation work, CMS has released multiple iterative updates to the methodology in December 2017, February 2019, and April 2021 that addressed concerns identified during stakeholder engagement activities such as the TEP and during public input.

The prior version of the methodology released in April 2021 was updated to improve the simplicity, predictability, and comparability of the Overall Star Rating methodology. Since then, a dry run was completed to support the inclusion of VHA hospitals in the Overall Star Rating beginning with the July 2023 Overall Star Rating release. Additionally, since the prior April 2021 Methodology Report, several stakeholder engagement efforts were convened focusing on reevaluation including four Provider Workgroup meetings, four Patient and Family Engagement Workgroup meetings, four public comment periods and one TEP.

At several of these stakeholder meetings, namely the 2023 and 2024 workgroup meetings and TEP, potential ways of emphasizing Safety of Care in the Overall Star Rating methodology were discussed. This stakeholder feedback combined with public comment periods and CMS's broader patient safety goals, culminated in the below update to the methodology implemented in v5.1.

- Application of a 4-star cap for hospitals in the lowest quartile of the Safety of Care measure group performance (and with at least three safety measure scores) starting in Calendar Year 2026.

The following update is anticipated to replace the 4-star cap beginning in Calendar Year 2027, however, as customary, updated methodology documentation will be published with the next release as appropriate.

- *Application of a blanket 1-star reduction for hospitals in the lowest quartile of the Safety of Care measure group performance (and with at least three safety measure scores) starting in Calendar Year 2027.*

This year's methodology change was proposed and codified in the federal Notice-and-Comment rulemaking process. The CY 2025 OPPS/ASC Proposed Rule³ (CMS-1809-P) included a Request for Information that included several options for emphasizing the Safety of Care measure group in the Overall Star Rating methodology where public comments were then responded to and addressed in the CY 2025 OPPS/ASC Final Rule⁵ (CMS-1809-FC). Then, in the subsequent CY 2026 OPPS/ASC Proposed Rule⁴ (CMS-1834-P), the 4-star cap for hospitals in the lowest quartile of the Safety of Care measure group beginning in 2026 and the 1-star reduction for hospitals in the lowest quartile of the Safety of Care measure group beginning in 2027 were formally proposed. These proposed methodology updates

were then finalized in the CY 2026 OPPS/ASC Final Rule⁶ (CMS-1834-FC) which can be found at [Federal Register :: Medicare Program: Hospital Outpatient Prospective Payment and Ambulatory Surgical Center Payment Systems; Quality Reporting Programs; Overall Hospital Quality Star Rating; Hospital Price Transparency; and Notice of Closure of a Teaching Hospital and Opportunity To Apply for Available Slots](#).

For questions regarding the Overall Star Rating Methodology, please use the [CMS Quality Question and Answer Tool](#).

Overall Star Rating Methodology

CMS considered various approaches to calculating the Overall Star Rating for the original design of the Overall Star Rating Methodology, including using simple or weighted averages of eligible measures, complex statistical approaches utilizing factor analysis and latent variable modeling, or using fixed cut points or clustering algorithms for assigning the final star rating. CMS evaluated each approach in the context of the project goals and stakeholder input.

CMS sought to identify an approach that would:

- Generate a single, aggregate measure of available hospital quality information;
- Account for the heterogeneity of measures available (process, outcome, etc.);
- Account for the fact that different hospitals report different numbers and types of measures;
- Accommodate changes in the included measures (for example, retirement of measures); and,
- Utilize an evidence-based approach reflecting modern statistical methods that previously have been applied to health care.

The methodology calculates the Overall Star Rating through a nine-step process ([Appendix B](#)). These steps are listed below and described in greater detail in subsequent sections.

- Step 1: Selection of measures for inclusion in the Overall Star Rating
- Step 2: Standardization of measures for direction and magnitude
- Step 3: Assignment of measures to groups
- Step 4: Calculation and standardization of measure group scores
- Step 5: Calculation of hospital summary scores as a weighted average of available group scores
- Step 6: Application of minimum thresholds for receiving an Overall Star Rating
- Step 7: Assignment of hospitals to peer groups based on their number of measure groups (3, 4, or 5)
- Step 8: Application of K-means clustering
- Step 9: Calculation of Overall Star Rating

The measures are first selected based on their relevance and importance as determined through stakeholder and expert feedback. The selected measures are standardized to be consistent in terms of direction and magnitude (Step 2). In Step 3, the measures are organized into five groups by measure type. In Step 4, a simple average of the standardized measure scores is calculated and this score is again standardized to generate a measure group score. In Step 5, a predetermined weight is applied to each measure group score to calculate a hospital summary score. In Step 6, the public reporting threshold is applied, requiring hospitals to have a minimum of three measure groups (one of which must be the Mortality or Safety of Care group) with at least three measures in each of the three groups to receive an Overall Star Rating. In Step 7, hospitals meeting the reporting thresholds are organized into peer groups based on the number of complete measure groups they have submitted. In Step 8, hospital summary scores are categorized into 5 clusters. In the newly added Step 9, 5 stars are assigned to the highest cluster and 1 star to the lowest cluster; furthermore a 4-star cap is applied to hospitals that score in the lowest quartile on their Safety of Care measure group score and report at least three safety measures.

For the 2026 Overall Star Rating, the only step of the methodology that is different is the newly added Step 9. All other previously established components of the methodology remain the same, however, we

note that Step 1 and Step 2 (selection and standardization of measures for inclusion) were previously coupled in a single Step 1; we have split these into distinct steps to distinguish more clearly between them and align with regulatory language.

The Overall Star Rating is calculated using R. In alignment with the key principle of transparency, the R code pack as well as the Overall Star Rating input file and corresponding technical documentation can be found on QualityNet.

Generally, the Overall Star Rating is updated yearly using data publicly reported from a quarter within the prior 12 months as finalized in the CY 2023 OPPS/ASC Final Rule⁷ (CMS-1772-FC). The April 2026 Overall Star Rating was calculated using measure data from the October 2025 refresh of [Care Compare on Medicare.gov](#).

Step 1: Measure Selection for Inclusion

Introduction to Care Compare Measures

Care Compare on Medicare.gov includes measures that reflect a range of different dimensions of quality, from clinical care processes, to measures focused on care transitions, to measures of patient experience. The measures reported via [Care Compare on Medicare.gov](#) represent a variety of measure types and cover a broad set of clinical conditions and care processes. Though not all measures reported via [Care Compare on Medicare.gov](#) are included in the Overall Star Rating, the Overall Star Rating includes a broad and diverse set of measures.

Criteria for Selecting Measures for the Overall Star Rating

CORE vetted measure selection criteria with stakeholders through the original TEP and public input periods to ensure that the Overall Star Rating captured the diverse aspects of quality represented by the measures reported via [Care Compare on Medicare.gov](#). CMS believes that these measures capture the quality of care at hospitals and appropriately provide a snapshot of quality when combined.

All measures for acute inpatient and outpatient care hospitals reported via [Care Compare on Medicare.gov](#) in October 2025, as determined using the data reported in the downloadable data file available on [data.cms.gov](#)[‡], are considered potentially eligible for inclusion in the Overall Star Rating. Any measures reported via [Care Compare on Medicare.gov](#) for other types of hospital (specialty hospitals such as cancer or inpatient psychiatric facilities) or ambulatory surgical centers are not eligible for inclusion.

Lastly, through discussion with CMS, the HCAHPS team and CMS contractors, CMS calculates Patient Experience group scores using linear mean scores for the HCAHPS measures as well as the OAS CAHPS measures.

Measure Exclusion Criteria

CMS uses the following criteria to exclude potential measures from the Overall Star Rating calculation:

[‡] Data is published to PDC for hospitals that have their Notice of Participation for IQR/OQR (<https://qualitynet.cms.gov/inpatient/iqr/participation#tab2>). The downloadable data file is available at <https://data.cms.gov/provider-data/search?theme=Hospitals>

1. Structural measures not amenable to inclusion in a summary scoring calculation alongside process and outcome measures, as these measures cannot be as easily combined with other measures captured on a continuous scale with more granular data;
2. Non-directional measures (for which it is unclear whether a higher or lower score is better), as these measures cannot be standardized to form an aggregate measure group score;
3. Measures published to Care Compare on Medicare.gov that are not required for public reporting (that is, for which reporting is voluntary or not mandated) through CMS programs,
4. Overlapping measures (for example, measures that are identical to another measure, or measures with substantial overlap in cohort and/or outcome), and
5. Measures with no more than 100 hospitals reporting performance publicly, as these measures would not produce reliable measure group scores based on so few hospitals.

With these measures excluded, a total of 52 measures met the criteria for inclusion in the April 2026 Overall Star Rating. The Overall Star Rating measure selection criteria are presented in the subsequent text and in [Figure 3](#).

Step 2: Measure Standardization

Standardization of Measure Scores

Hospital quality measure results include many different types of scoring information, ranging from time (e.g., median time in minutes from emergency department [ED] arrival to ED departure for admitted ED patients) to percentages (e.g., percentage of patients who received appropriate care for severe sepsis and septic shock). The included quality measures can also have two directions, with either “lower is better” (readmissions, mortality) or “higher is better” (percent of healthcare workers vaccinated against Influenza). Therefore, to facilitate the combination of information, CMS uses standardization to ensure all measure scores are on a common scale with a common direction. This does not change the measure information, just the scale for scoring, and makes it possible to combine the measures in the Overall Star Rating calculation so that all measures have the same weight. Specifically, CMS standardizes a hospital’s score on each measure by calculating “Z-scores” for each measure, reversing the direction if necessary, so that higher values are always ‘better’; the measure “Z-score” is the difference between an individual hospital’s score and the overall mean score for hospitals divided by the standard deviation across hospitals.

For example, in April 2026, Patient Safety Indicator (PSI) 4 (Death Among Surgical Patients with Serious Treatable Complications) had a national average performance of 174 deaths with a standard deviation of 23 deaths. After standardization and redirection, this measure had a mean score of 0 and standard deviation of 1 and the score was reversed so that a higher standardized score indicated better quality (fewer deaths). Henceforth in this report, a measure score refers to the standardized measure score or “Z-score”.

Step 3: Assignment of Measures to Groups

Approach to Grouping Measures

The Overall Star Rating measure groups as of April 2026 are:

1. Mortality;
2. Safety of Care;
3. Readmission;
4. Patient Experience; and,
5. Timely and Effective Care.

The rationale for these five measure groups is as follows:

- The five groups are aligned with national quality initiatives, such as the CMS Hospital Value-Based Purchasing (HVBP) program;
- The groups are clinically reasonable in that they capture common aspects of quality for which hospital quality is likely linked across measures; and,
- The groups make it easier for future measures related to these groupings to be added or removed from the Overall Star Rating as they are updated through [Care Compare on Medicare.gov](https://carecompare.cms.gov).

The group names were finalized with input from the TEP and previous CMS consumer testing.

For the specific measures in each group as of April 2026, please refer to [Overall Star Rating Methodology Results \(April 2026\)](#).

Step 4: Calculation of Measure Group Scores using a Simple Average of Measure Scores

Calculating Measure Group Scores Using a Simple Average of Standardized Measure Scores

To calculate the measure group score, the standardized measure scores for each measure in a group, calculated in Step 2, are averaged. In other words, each measure is given equal weight in the measure group score calculation. This approach to calculating the measure group scores increases the simplicity and predictability of measure weights within the remaining steps of the methodology. It was supported by the TEP and other stakeholders who favored a less complex methodology that could be easily understood by stakeholders and used by hospitals to identify areas for quality improvement.

To calculate the measure group score, 100 percent is divided by the number of measures reported to give the percentage each measure will weigh for that hospital; this measure weight is then multiplied by the standardized measure score to calculate the measure's weighted score. Then, the weighted individual measure scores within a group are added together to calculate the measure group score. If a hospital reports no measures for a given measure group, CMS considers that group to be "missing." See [Table 1](#) below for an example measure group score calculation.

Table 1. Example Measure Group Score Calculation

Measure	Hospital's Standardized Measure Score	Measure Weight
MORT-AMI	0.5	1 (17%)
MORT-CABG	n/a	-
MORT-COPD	0.5	1 (17%)
MORT-HF	-0.7	1 (17%)
MORT-30-PN	0.7	1 (17%)
MORT-30-STK	n/a	-
Hybrid-HWM (MORT-30-HOSPWIDE)	-0.3	1 (17%)
PSI-4	0.8	1 (17%)

Group Score = $\text{sum}(\text{measure scores} * \text{measure weights}) =$
 $(0.5 * .17) + (-0.7 * .17) + (0.5 * .17) + (0.7 * .17) + (-0.3 * .17) + (0.8 * .17) = \mathbf{+0.25}$

As noted in [Step 6](#) below, hospitals must report at least three measures within at least three measure groups, one of which must be Mortality or Safety of Care, to receive an Overall Star Rating. As soon as this threshold is met, all available measures and groups will contribute to that hospital's Overall Star Rating, including any groups with just one or two measures.

The contribution of each individual measure to its respective measure group score can vary, depending on how many measures in that group are reported. Thus, measure emphasis will vary across hospitals and across groups within a hospital. As an example, if a hospital reports three measures in the Safety of Care measure group, each measure would contribute 1/3 (33%) to the group score, whereas if a hospital reports all eight measures each would contribute 1/8 (12.5%) to the group score.

See [Table 2](#) below for an example of how measures would be combined through a simple average of measure scores to calculate measure group scores.

Table 2. Example Calculation of Measure Group Score

Measure Name	Example Measure Score	Standardized Measure Score	Measure Weight	Weighted Standardized Measure Scores*	Safety of Care Measure Group Score
COMP-HIP-KNEE Hospital-Level Risk Standardized Complication Rate (RSCR) Following Elective Primary Total Hip Arthroplasty (THA) and Total Knee Arthroplasty (TKA)	3.22%	-1.13	12.5%	-0.14	0.23
HAI-1 Central-Line Associated Bloodstream Infection (CLABSI)	1.233	-0.75	12.5%	-0.09	
HAI-2 Catheter Associated Urinary Tract Infection (CAUTI)	0.747	0.09	12.5%	0.01	
HAI-3 Surgical Site Infection (SSI) from Colon Surgery	0.000	1.21	12.5%	0.15	
HAI-4 Surgical Site Infection (SSI) Abdominal Hysterectomy	0.000	0.97	12.5%	0.12	
HAI-5 Methicillin Resistant Staphylococcus Aureus (MRSA) Bacteremia	0.166	0.98	12.5%	0.12	
HAI-6 Clostridium Difficile (C. difficile)	0.470	0.46	12.5%	0.06	
CMS PSI-90 Patient Safety and Adverse Events Composite	0.999	0.02	12.5%	0.003	

*Measure group scores are continuous and rescaled. Negative and positive scores do not denote good or bad performance

Standardization of Measure Group Scores

As with the individual measures, standardizing scores is a way to facilitate combining them into a final overall score by ensuring they are scaled on a comparable basis. Using the simple average of measure group scores the distributions and interpretations of measure group scores may differ. For example, a 0.5 measure group score in Safety of Care may represent only 1 standard deviation above the mean, whereas in Patient Experience a 0.5 measure group may represent 2 standard deviations above the mean. Standardizing the group scores ensures that similar values represent similar performance relative to all others.

Specifically, measure scores are standardized by calculating a Z-score for each measure group. The Z-score is calculated by subtracting the national average measure group score from each hospital's

measure group score and dividing by the standard deviation across hospitals. This step occurs prior to combining measure group scores through a weighted average to calculate summary scores.

Step 5: Weighted Average of Measure Groups to Calculate Summary Scores

Approach to Developing the Weighting Scheme

After estimating the measure group score for each hospital and each group, CMS calculates a weighted average to combine the five measure group scores into a single hospital summary score. CMS evaluated potential weighting options considering the following criteria:

1. Measure Group Importance
 - The weight of Mortality, Safety of Care, Readmission, and Patient Experience (which include measures of outcomes) should be greater than that of Timely and Effective Care (which comprises process of care measures).
2. Stakeholder Input
 - The weights should reflect the prioritization of the measure groups by the TEP as well as feedback received during the public input periods, the Overall Star Rating dry run, and additional sources of patient and consumer feedback.

Final Weighting Scheme

After calculating the measure group scores for each hospital, CMS calculates a weighted average to combine the measure group scores into a single hospital summary score. The final weighting scheme was developed in collaboration with various stakeholder groups. Stakeholders agreed that the Mortality, Safety of Care, Readmission and Patient Experience measure groups were more indicative of quality and, thus, should be weighed heavier than the Timely and Effective measure group. The weights were reviewed through the TEP, Provider WG, PFE WG, and public input in 2017, where the majority of feedback warranted no change to the original weighting scheme. These weights were again reassessed in 2019 across an array of stakeholders and public comment to account for the consolidation of all process measures (formerly comprising three measure groups) into a single Timely and Effective Care Measure Group. The weights for each Measure Group are detailed in [Table 3](#).

Table 3. Overall Star Ratings Weighting by Group

Group	Star Ratings Weight (w_d)
Mortality	22%
Safety of Care	22%
Readmission	22%
Patient Experience	22%
Timely and Effective Care	12%

Method for Re-weighting When Missing One or more Measure Groups

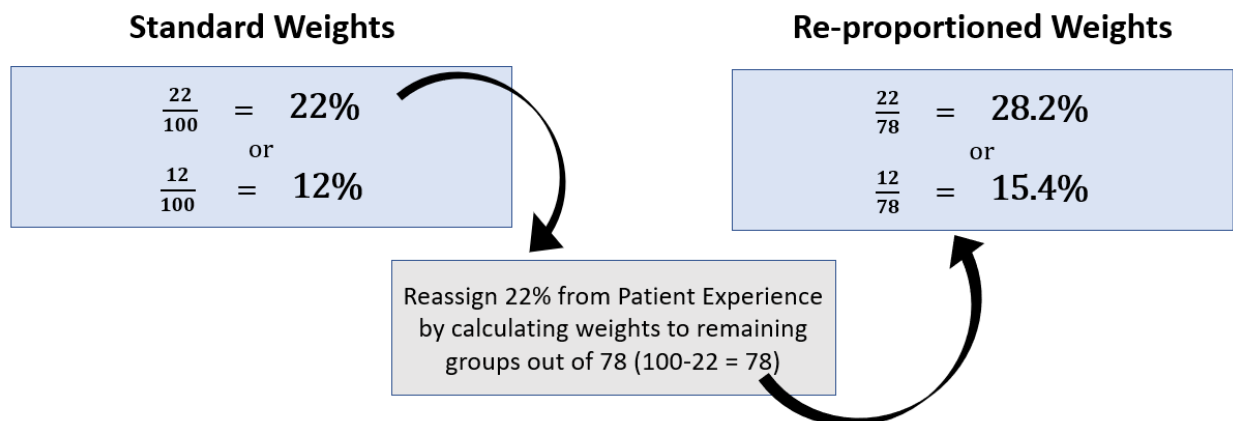
If a hospital reports no measures for a given measure group, CMS considers that group to be “missing.” When a hospital is missing one or more measure groups, CMS applies the HVBP’s approach of re-proportioning the weight of the missing group(s) across the groups for which the hospital does report measures. [Table 4](#) and [Figure 1](#) provide examples of how the weighting scheme is adjusted for a hospital that is missing the Patient Experience measure group.

The final summary score for each hospital is the weighted average of that hospital’s measure group scores.

Table 4. Example Re-weighting Scheme for Hospital Missing Patient Experience Group

Measure Group	# measures in group	Standard Weight	Re-proportioned Weight
Mortality	4	22%	28.2%
Safety of Care	2	22%	28.2%
Readmission	3	22%	28.2%
Patient Experience (missing)	0	22%	-
Timely and Effective Care	6	12%	15.4%

Figure 1. Example Calculation for Re-proportioning Measure Group Weights



Step 6: Minimum Thresholds for Reporting an Overall Star Rating

Requirements for Receiving an Overall Star Rating

CMS aims to assign Overall Star Ratings on the basis of adequate information regarding hospitals' quality. Thus, CMS evaluated and developed standards regarding the minimum number of measures and groups a hospital must report to receive a publicly reported Overall Star Rating. CMS set these thresholds to allow for as many hospitals as possible to receive an Overall Star Rating without sacrificing the validity and reliability of the Overall Star Rating methodology. This step is applied prior to clustering hospitals into star categories.

In the past, hospitals that provide acute inpatient and outpatient care were required to publicly report a sufficient number of measures to receive an Overall Star Rating. CMS sets a minimum threshold to require at least three measure groups with at least three measures in each, one of which must be Safety of Care or Mortality. **As long as a hospital meets this threshold, all available measures and measure groups are incorporated in their summary score (that is, even measure groups with only 1 or 2 measures would be included).** While this results in instances in which a hospital has one or two measures in a group that contribute a substantial portion of a hospital's summary score, incorporating all measures for which a hospital has scores aligns with one of the guiding principles of inclusivity of measure information. This approach was vetted through a TEP and public comment stakeholder engagement⁸ and allows as many hospitals as possible to receive an Overall Star Rating while also maintaining face validity and reliability of the Overall Star Rating methodology.

Step 7: Application of Peer Grouping by Number of Measure Groups with at Least Three Measures

Background

In early versions of the Overall Star Rating methodology, CMS did not previously group hospitals by number of measures or measure groups. The decision to peer group hospitals by the number of measure groups reported is responsive to stakeholder concerns about the comparability of hospital ratings, allowing hospitals to be assigned star ratings relative to other hospitals with similar quality measure information. Hospitals have been identified to report different numbers and types of measures based on their size, patient volume, services provided (service mix), or patients treated (case mix). For example, larger hospitals with more diverse case mix and service mix, such as large urban teaching hospitals, frequently report a greater number of measures, and therefore measure groups. This places them into a different peer group than smaller hospitals with less diverse patient cases and service mix, which tend to report fewer measures and measure groups.

Peer grouping as a final step before the assignment of hospitals to an Overall Star Rating allows:

- (1) Hospital summary scores to be equivalent and comparable among all hospitals, regardless of peer grouping;
- (2) Transparency and the ability for hospitals to review measure group and summary score results comparable to all other hospitals in the nation within their confidential hospital-specific reports during the 30-day confidential preview period upon public release, and the public (researchers

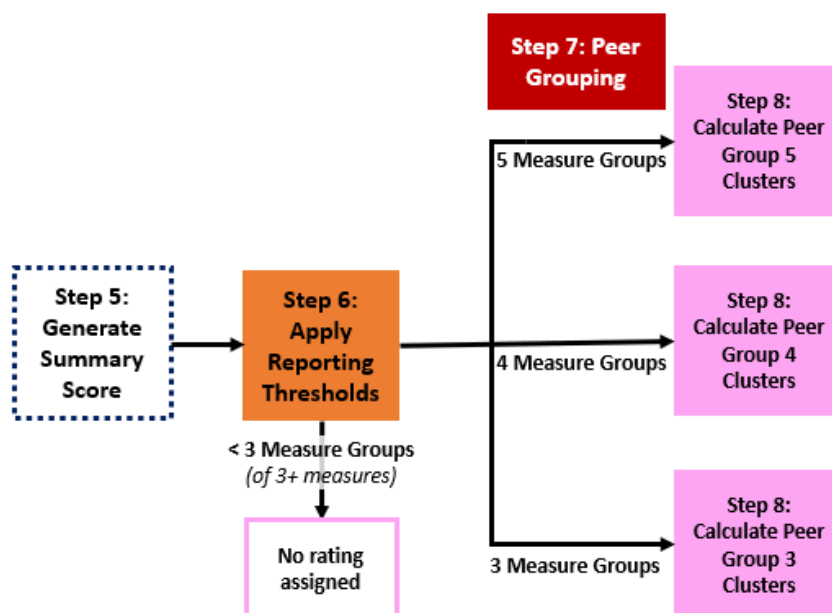
and patients) to review comprehensive data available through the downloadable data file on data.cms.gov;

- (3) Minimal sensitivity of measure-level differences between peer groups on their Overall Star Rating; and
- (4) Comparisons of Overall Star Ratings to hospitals that have a similar number of measure groups.

Approach to Peer Grouping

CMS peer groups hospitals by the number of measure groups for which they have at least three measures. Specifically, after the minimum reporting thresholds are applied, hospitals are categorized into one of three peer groups: a 3-measure group peer group (Peer Group 3), a 4-measure group peer group (Peer Group 4), and a 5-measure group peer group (Peer Group 5). By definition, any hospital that qualifies for an Overall Star Rating in Step 6 is assigned to one of these peer groups, while any hospital that does not qualify in Step 6 receives no peer group assignment and no Overall Star Rating. K-means clustering (Step 8) is then applied within each peer group to partition hospital summary scores into discrete clusters, as shown in [Figure 2](#).

Figure 2. Approach to Peer Grouping



Step 8: Application of K-Means Clustering

Overview of k-Means Clustering

The Overall Star Rating methodology utilizes *k*-means clustering to classify hospitals with a continuous distribution of overall summary scores into discrete categories for Overall Star Rating assignment. *K*-means clustering is implemented within each peer group (peer group 3, peer group 4, peer group 5) separately, so hospitals are only compared to similar facilities. The reporting threshold (Step 6) and peer group assignment (Step 7) are both applied immediately prior to *k*-means clustering (Step 8) to ensure that summary score cut points and each hospital's cluster are determined by comparison only to other hospitals within the same peer group.

The *k*-means clustering analysis is a standard method for creating categories (or clusters) so that the observations (or scores) in each category are closer to their category mean than to any other category mean.⁹ The number of categories is pre-specified; CMS has specified five categories, so that the *k*-means clustering analysis generates five categories based on hospital summary scores in a way that minimizes the distance between summary scores and their assigned cluster mean. Stated in another way, hospitals were organized into one of five categories such that a hospital's summary score is "more like" that of the other hospitals within the same category and "less like" the summary scores of hospitals in the other categories. The methodology runs clustering until complete convergence, in which the procedure iteratively examines solutions until it can find no more optimal solution.

The rationale for the decision to use *k*-means clustering is as follows:

- *K*-means optimally designates five "means" for five clusters within the distribution of hospital summary scores in a way that minimizes the within-category and maximizes the between-category differences in summary scores.
- Hospitals in a cluster will have similar summary scores.
- This approach is aligned with the similar clustering approach used to calculate the HCAHPS Star Ratings, also reported via Care Compare on Medicare.gov.

Results of CMS's analyses of the validity and the reliability of this approach using 2021 Overall Star Rating results are shown in [Appendix F](#).

Step 9: Calculate Star Rating

Assumptions for Translating Summary Scores to Stars

The final Overall Star Rating categories are structured such that the lowest hospital summary score cluster is assigned one star, and the highest cluster is assigned five stars, assigned separately for each peer group. Prior to selecting an approach for translating hospital summary score categories to stars, CMS considered several important assumptions:

- Any approach selected will always result in some hospitals having summary scores at the boundary of two star categories (that is, some hospitals will closely border a higher or lower star category).
- Similar to other CMS star rating efforts, a three-star rating will be considered "average."

- The objective of this project is to publish whole-star ratings (not half-stars).
- The Overall Star Rating does not reflect an “apples to apples” comparison between hospitals – in other words, just because two hospitals may have the same Overall Star Rating does not mean they have identical hospital quality. Rather, the Overall Star Rating reflects the weighted average of the summarized quality information available for a given hospital.
 - For example, there are many ways a hospital can achieve a 3-star rating. One hospital may do exceedingly well on the Mortality and Readmission groups but perform poorly on Patient Experience. Another hospital with the same rating may have average performance across all available measure groups.
- Because each hospital may have a different set of measures contributing to its Overall Star Rating, patients and consumers should evaluate individual measure scores in addition to the Overall Star Rating. Individual measure performance can be found for a given hospital via [Care Compare on Medicare.gov](https://carecompare.medicare.gov).
- The Overall Star Rating is not intended to guide specific hospital quality improvement efforts, but rather to make summary information available to the public.

Safety of Care Cap

Given CMS’s priority to emphasize patient safety across the Department of Health & Human Services as part of broader federal efforts, this component of the methodology intends to align the Overall Star Rating with broader patient safety goals nationally. Therefore, a “safety cap” is applied in the final step of the Overall Star Rating methodology when the Overall Star Rating is calculated to further emphasize the Safety of Care measure group. Any hospital with at least three Safety of Care measures and a lowest-quartile Safety of Care score is eligible to receive a maximum rating of four stars.

First, hospitals are categorized into performance quartiles based on their Safety of Care measure group scores. Quartiles are calculated from the distribution of all hospitals with at least one Safety of Care measure, whether or not they qualify for an Overall Star Rating, which is in alignment with the guiding principle of the Overall Star Rating of inclusiveness of hospital and measure information.

Second, Overall Star Ratings are assigned based on the k-means clusters for each peer group (Step 8), with those in the lowest-scoring cluster assigned one star and those in the highest-score cluster assigned five, with the exception that any hospital within the highest-scoring cluster but subject to the cap (that is, at least three Safety of Care measures and lowest-quartile Safety of Care score) is instead assigned four stars accordingly. Hospitals with just one or two Safety measures are not subject to the cap regardless of Safety group score quartile given the lack of stability and precision in measure group scoring with limited measure availability.

Given that patient safety constitutes a fundamental component of the CMS National Quality Strategy, this final step of the Overall Star Rating methodology reflects CMS program alignment in striving for optimal health outcomes.

Updates to the 2026 Overall Star Rating

Some updates to individual measures included in the Overall Star Rating were made to data published to [Care Compare on Medicare.gov](https://carecompare.medicare.gov) since the last Overall Star Rating refresh, which are reflected in the April 2026 refresh.

One measure previously included in the Overall Star Rating was retired from public reporting in January 2025 and is no longer included as of the 2026 rating:

- PC-01: Percent of newborns whose deliveries were scheduled too early (1-3 weeks early), when a scheduled delivery was not medically necessary

One measure previously included in the Overall Star Rating was retired from IQR in July 2025 and from OQR in October 2025:

- HCP COVID-19: COVID-19 Vaccination Coverage Among HCPs

In addition, several new measures were added to reporting on Care Compare on Medicare.gov since the last Overall Star Rating refresh, six of which meet the criteria for the Overall Star Rating and will be included beginning in April 2026:

- Hybrid HWM (Mort-30-HOSPWIDE): Death rate for patients (hospital-wide)
- O-COMP-1 Patients that reported if staff gave care in a professional way and if the facility clean
- O-COMP-2: Patients who reported if they received communication about what to expect during and after the procedure
- O-COMP-3: Patients who reported if they were prepared for discharge and recovery
- O-PATIENT-RATE: Patients who gave the facility a rating
- O-PATIENT-REC: Patients who reported if they would recommend the facility

The following measures were also recently added to public reporting on Care Compare on Medicare.gov and are potentially eligible for inclusion for this reason, but satisfy one of the Overall Star Rating exclusion criteria (step 1) and therefore will not factor into hospitals' Overall Star Rating:

- OP-18a: Average (median) time all patients spend in the emergency department before leaving from the visit
 - Rationale: Substantial overlap in cohort & outcome with OP-18b
- OP-18c: Average (median) time psychiatric or other mental health patients spent in the emergency department before leaving from the visit
 - Rationale: Non-directional measure
- OP-18d: Average (median) time transfer patients spent in the emergency department before leaving from the visit
 - Rationale: Non-directional measure

Overall Star Rating Methodology Results (April 2026)

This section describes results of the Overall Star Rating methodology, using October 2025 [Care Compare on Medicare.gov](https://carecompare.medicare.gov) data for the April 2026 publication of the Overall Star Rating. CMS presents the national distribution of the Overall Star Rating and results for each step of the methodology (measure selection, assignment to measure groups, measure group scores, summary scores, and star classification) via the [Care Compare on Medicare.gov](https://carecompare.medicare.gov) website.

Distribution of Overall Hospital Star Ratings

Together, both the minimum measure and group thresholds resulted in 70.1 percent of hospitals (N=3,203) receiving an April Overall Star Rating using October 2025 data. Since the initial public reporting in July 2016, between 61.1 and 80.3 percent of hospitals have received an Overall Star Rating each reporting period. Setting higher thresholds for both measures and groups would exclude more hospitals from the Overall Star Rating.

The frequency of hospitals by each star rating category is shown in [Table 5](#). Similar to prior publications of the Overall Star Rating[§], in April 2026 there were more two-, three-, and four-star hospitals and fewer one- and five-star hospitals.

Table 5. Distribution of Overall Star Ratings (n=3,203)

Overall Star Rating	Number of Hospitals (% Total)
★★★★★	385 (12.0%)
★★★★	953 (29.8%)
★★★	991 (30.9%)
★★	670 (20.9%)
★	204 (6.4%)

Measure Selection

[Figure 3](#) depicts the process of measure selection for April 2026. Out of 64 measures eligible for inclusion on Care Compare, 12 measures were excluded based on our selection criteria. See [Appendix C](#) for a list of excluded measures.

Assignment to Groups

[Table 6](#) displays the number of measures assigned to each measure group included in the Overall Star Rating for April 2026. The complete list of measures by group can be found in [Appendix D](#).

[§] When compared to July 2025 Overall Star Ratings distribution, available in the July 2025 Updates and Specifications Report. <https://qualitynet.cms.gov/inpatient/public-reporting/overall-ratings/resources#tab2>

Figure 3. Measure Selection Flowchart (April 2026 Overall Star Rating, October 2025 Data)

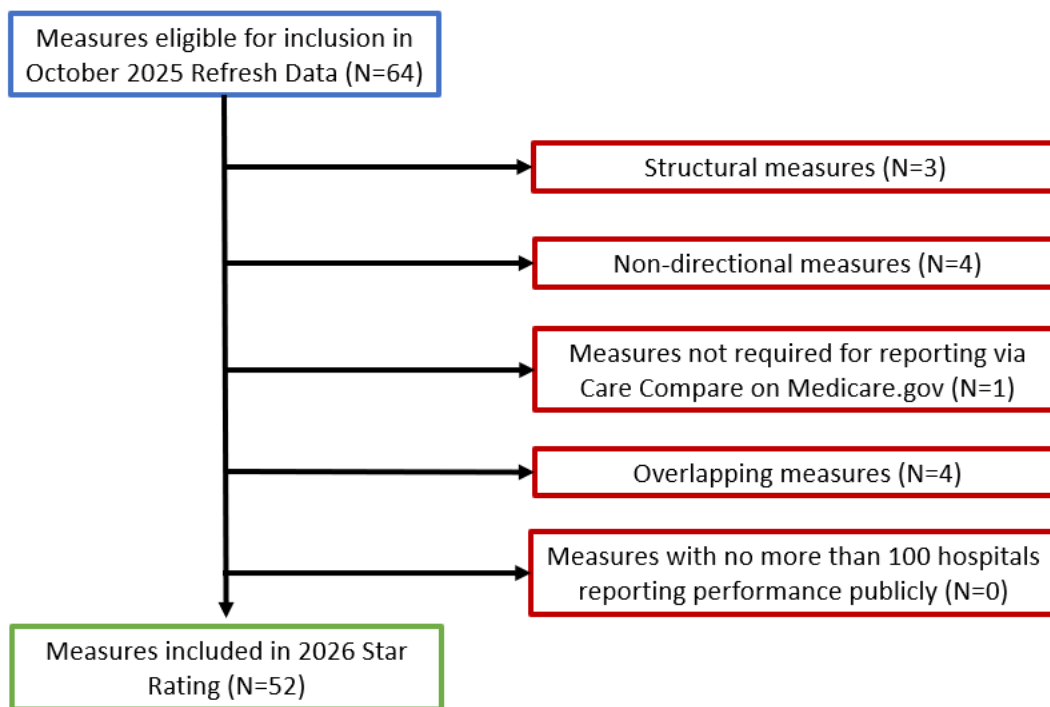


Table 6. Total Number of Measures by Measure Group

Group	Number of Measures (N=52)
Mortality	8
Safety of Care	8
Readmission	11
Patient Experience	15
Timely & Effective Care	10

Measure Group Scores

Measure Group Performance Category Results for April 2026

In version 4.1 of the methodology, we provided more information and results related to measure group performance, details of which can be found in [Appendix E](#).

[Table 7](#) displays the distribution of hospital group scores in each measure group for the April 2026 Overall Star Rating, using [Care Compare on Medicare.gov](#) data from October 2025. Note that group scores are defined such that a higher group score indicates better performance on the underlying measures.

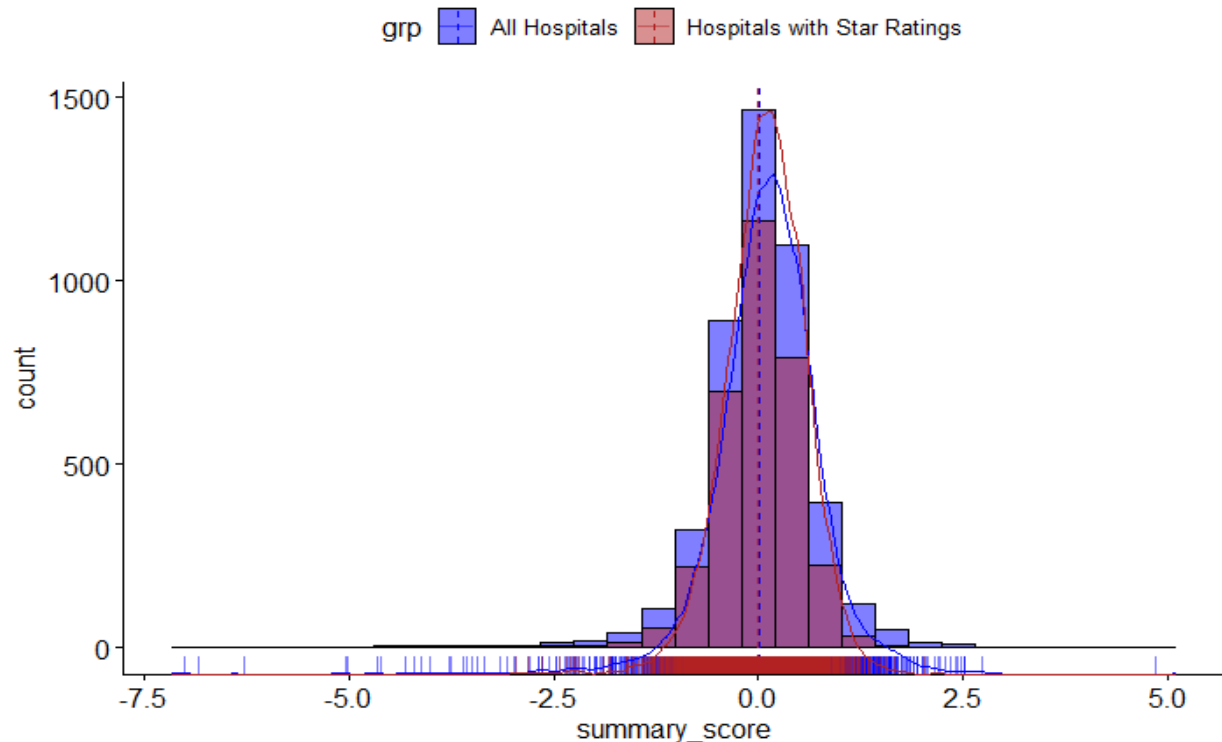
Table 7. Distribution of Measure Group Score (among hospitals with at least one measure in a given group)

Group	National Average Score	Standard Deviation	Minimum	25 th Percentile	Median	75 th Percentile	Maximum
Mortality (N=4133)	-0.079	0.7382	-6.404	-0.498	-0.030	0.392	3.014
Safety of Care (N=3372)	-0.024	0.8335	-13.566	-0.277	0.087	0.412	2.354
Readmission (N=4308)	0.000	0.5635	-4.341	-0.293	0.037	0.346	2.784
Patient Experience (N=3496)	0.027	0.8383	-10.563	-0.430	0.072	0.557	2.967
Timely and Effective Care (N= 4549)	0.017	0.5126	-5.263	-0.231	0.048	0.323	2.678

Hospital Summary Scores

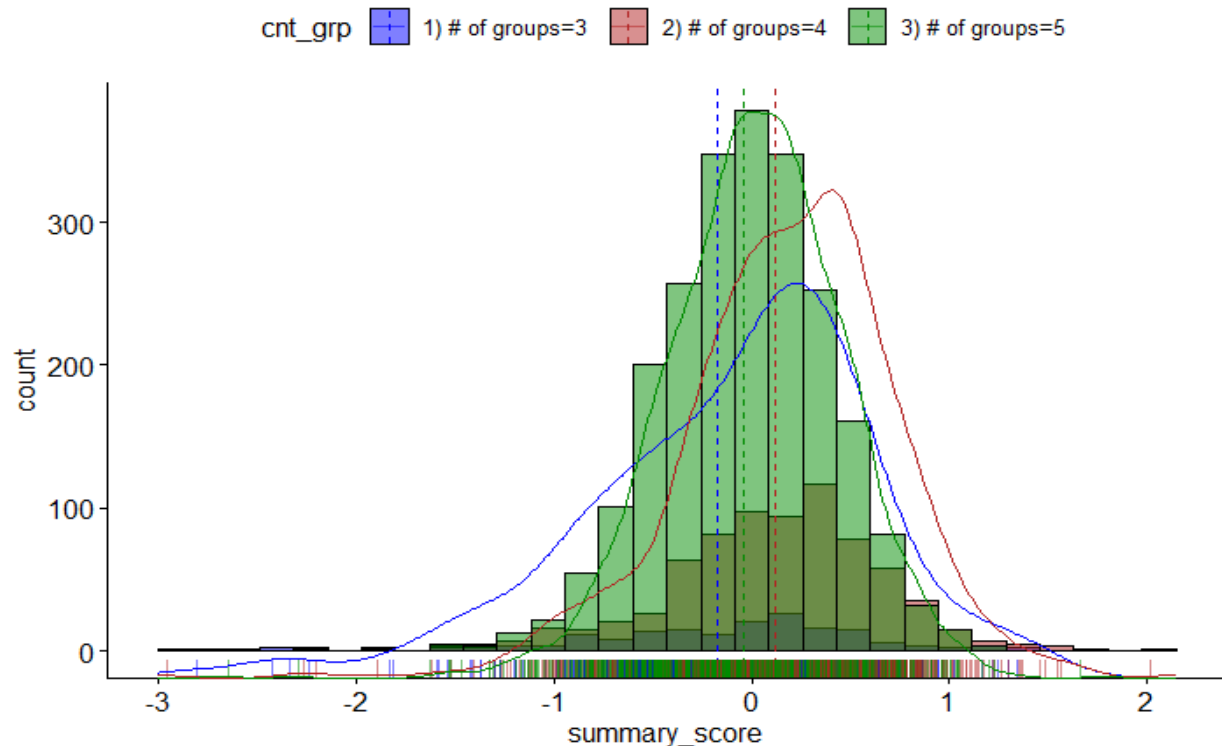
[Figure 4](#) presents the distribution of hospital summary scores for April 2026. The bars represent the percentage of hospitals (y-axis) with a given hospital summary score (x-axis). Of 4,569 hospitals that received a summary score, 3,203 met the criteria for reporting of an Overall Star Rating.

Figure 4. Distribution of Hospital Summary Scores for April 2026: All Hospitals (n=4,569, in blue), and Hospitals receiving an Overall Star Rating (n=3,203, in red)



[Figure 5](#) similarly presents the distribution of hospitals' summary scores for each of the three peer groups. Of the 3,203 hospitals meeting the criteria to receive an Overall Star Rating, 177 were in Peer Group 3 (in blue), 749 were in Peer Group 4 (in red), and 2,277 were in Peer Group 5 (in green).

Figure 5. Distribution of Hospital Summary Scores for April 2026: Peer Grouped (3-group, n=177; 4-group, n=749; 5-group, n=2,277)



Star Classification

[Table 8](#) shows the range of summary scores for each star category cluster within each peer group.

Table 8. K-means Overall Star Rating Summary Score Cluster Ranges by Peer Group*

Rating	3-measure group peer group (n= 177)	4-measure group peer group (n= 749)	5-measure group peer group (n= 2277)
1 Star	-2.808, -1.276	-2.954, -0.745	-2.648, -0.694
2 Star	-1.201, -0.466	-0.730, -0.164	-0.691, -0.277
3 Star	-0.439, 0.085	-0.160, 0.244	-0.276, 0.063
4 Star	0.104, 0.645	0.247, 0.687	0.063, 0.423
5 Star	0.678, 1.339	0.703, 2.016	0.425, 1.656**

*Shows results for the 3,203 hospitals that met the reporting criteria.

The frequency of hospitals by each Overall Star Rating category for each peer group is shown in [Table 9](#).

** After summary score range determination, 15 hospitals in the 5-star cluster were assigned to the 4-star cluster in the final step of the Overall Star Rating methodology if in the bottom quartile Safety with at least three Safety measures. The summary score range among capped hospitals was 0.430-1.224.

Table 9. Distribution of hospitals by Overall Star Rating and Peer Group

Rating	3-measure group peer group (n= 177)	4-measure group peer group (n= 749)	5-measure group peer group (n= 2277)	Total (n=3203)
1 Star	15 (8.5%)	49 (6.5%)	140 (6.1%)	204 (6.4%)
2 Star	39 (22.0%)	147 (19.6%)	484 (21.3%)	670 (20.9%)
3 Star	49 (27.7%)	232 (31.0%)	710 (31.2%)	991 (30.9%)
4 Star	60 (33.9%)	235 (31.4%)	658 (28.9%)	953 (29.8%)
5 Star	14 (7.9%)	86 (11.5%)	285 (12.5%)	385 (12.0%)

Safety of Care Results

Under version 4.1 of the methodology in the July 2025 Overall Star Rating, 14 hospitals (with at least three Safety of Care measures) had received a 5-star rating despite bottom quartile Safety of Care performance. The methodology update (v5.1) addresses this rare but meaningful occurrence. In the 2026 Overall Star Rating, 15 hospitals' ratings were impacted by the Safety of Care cap, and no such hospitals received a 5-star rating.

The Safety of Care measure group score has the potential for a varied percentage contribution to a hospital's summary score. Specifically, there is a broader range and higher median among Peer Group 3 hospitals compared to Peer Group 4 and 5 hospitals as there are fewer measures and measure group scores available toward calculating the hospital's summary score. As shown in [Table 10](#), no extreme weightings of the Safety of Care measure group contribution were observed, with these findings more a function of proportional re-weighting ([Step 5](#)) is performed as opposed to the Safety of Care emphasis methodology update.

Table 10. Safety of Care Measure Group Weighting

Peer Group	N	Median (IQR) % Safety Measure Group Weighting Contributes to Hospital's Summary Score	Min, Max of Safety Measure Group Weighting Contributes to Hospital's Summary Score
3	177	0 (0, 22%)	0, 39%
4	749	22% (22%, 22%)	0, 28%
5*	2277	22% (22%, 22%)	22%, 22%
All	3203	22% (22%, 22%)	0, 39%

*By definition of peer grouping, all peer group 5 hospitals have all 5 measure groups at the base weight of 22% (12% for Timely & Effective Care).

Validity & Reliability of the Overall Hospital Star Rating Methodology

In version 4.1 of the Overall Star Rating methodology, we conducted extensive validity and reliability testing that demonstrated the simplification of the methodology upheld expected performance trends, details of which are outlined in [Appendix F](#).

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Appendix A: Stakeholder Roster

Table A - 1. November 2021 Provider WG Roster

Provider WG Member	Title
Eric Alper, MD, FACP, SFHM, CPE	UMass Memorial Health Care (<i>Chief Quality Officer and Chief Clinical Informatics Officer, Physician</i>)
Amy Aronsky, DO, MBA, FCCP, FAASM	Princeton Junction (<i>Senior Medical Director of Clinical Operations</i>)
Leah Berg, MS, RN, CPHQ	St. Luke's Hospital (<i>Quality Improvement Leader</i>)
Marjorie Bessel, MD	Banner Health (<i>Chief Clinical Officer</i>)
Maureen Dailey, PhD, RN, CWOON, FAAN	Daileysolution (<i>Consultant in Healthcare Quality</i>)
Kathleen DeSnyder, BS, MPH	Beaumont Health (<i>Senior Director of External Quality Measures</i>)
Suparna Dutta, MD, MS, MPH, FACP, FHM,	Rush University (<i>Associate Professor, Division of Hospital Medicine</i>)
Mark Fontana, PhD, MA, M.Eng.	Hospital for Special Surgery (<i>Senior Director of Data Science</i>)
Assad Ghani, BS, MBA, MHSA	UChicago Medicine (<i>Director of Value-Based Payment</i>)
Joy Gill, RN, MBA, CPHQ	Adventist HealthCare, Inc (<i>Director of Quality Regulatory Programs and Analytics</i>)
Karen Heller, MBA	Greater New York Hospital Association (<i>Senior Advisor of Health Economics and Outcomes Research</i>)
Amy Herlihy, RHIT, CCS	CommonSpirit Health (<i>Quality Analytics Coordinator</i>)
Linda Hightower, RN, ONC	Community Medical Center (<i>Quality Coordinator</i>)
Daniel Hoody, MD, MSc	Hennepin Healthcare (<i>Chief Medical Quality Officer</i>)
Sandi Hyde, BSME, MSPS	LifePoint Health (<i>Director of Quality Data</i>)
Kacie Kleja, MBA, MS	HCA Healthcare (<i>Assistant Vice President for Clinical Analytics and Public Reporting</i>)
John Krueger, MD, MPH	Common Spirit Health (<i>Vice President of Quality</i>)
Jennifer Lamprecht, MS, RN, CNL, CPHQ	Sanford Health (<i>Director of Quality Strategy</i>)
Brian Lewis, MBA	University Health System (<i>Vice President of Quality and Clinical Outcomes</i>)
Brian McCardel, MD	Sparrow Health System (<i>Surgeon, Board Quality Committee Chair</i>)
Amy Minnich, RN, MHSA	Geisinger Health System (<i>Director of Quality and Safety</i>)

Provider WG Member	Title
Katlyn Morin, MBA	The Carle Foundation Hospital (<i>Quality Data Analyst</i>)
Arby Nahapetian, MD, MPH, FACC	Adventist Health (<i>Chief Medical Officer</i>)
Henry Pitt, MD	Rutgers Cancer Institute of New Jersey (<i>Chief of Oncologic Quality</i>)
Martha Radford, MD, FACC, FAHA	NYU Langone Health (<i>Chief Quality Officer</i>)
Leslie Safier, MPH	Zuckerberg San Francisco General Hospital and Trauma Center (<i>Director of Performance Improvement</i>)
Georgia Thomas, RN, BSN, MBA, HACCP, CPHQ, FACHE	Memorial Hermann Health Systems (<i>System Associate Vice President of Quality</i>)
Ugochukwu Uwaoma, MD, MBA, MPH, FACP	Multicare Health System (<i>Physician and Chief Medical Officer</i>)
Julie Wall, RN, MBA, FACMPE	Benefis Health System (<i>Senior Vice President, Quality</i>)
Ronald Wyatt, MD, MHA	Cook County Health System (<i>Chief Quality Officer</i>)

Table A - 2. November 2021 PFE WG Roster

PFE WG Member
David Andrews
Michelle Beck
Anna Schwamlein Howard
Claire Noel-Miller
Ann Monroe
Linda Relyea
David Andrews
Michelle Beck

Table A - 3. January 2023 Provider WG Roster

Provider WG Member	Title
Eric Alper, MD, FACP, SFHM, CPE	UMass Memorial Health Care (<i>Chief Quality Officer and Chief Clinical Informatics Officer, Physician</i>)
Amy Aronsky, DO, MBA, FCCP, FAASM	United Health Care (<i>Senior Medical Director</i>)
Leah Berg, MS, RN, CPHQ	St. Luke's Hospital (<i>Quality Improvement Leader</i>)
Marjorie Bessel, MD	Banner Health (<i>Chief Clinical Officer</i>)
Maureen Dailey, PhD, RN, CWOON, FAAN	Daileysolution (<i>Consultant in Healthcare Quality</i>)
Kathleen DeSnyder, BS, MPH	Corewell Health (<i>Senior Director, Data Analytics & QA</i>)
Suparna Dutta, MD, MS, MPH, FACP, SFHM	Hartford Hospital (<i>Chair, Department of Medicine</i>)

Provider WG Member	Title
Mark Fontana, PhD, MA, M.Eng.	Hospital for Special Surgery (<i>Senior Director of Data Science</i>)
Assad Ghani, BS, MBA, MHSA	UChicago Medicine (<i>Director of Value-Based Payment</i>)
Joy Gill, DHA, RN, MBA, CPHQ	Sheppard Pratt (<i>Clinical Director, Quality and Safety</i>)
William Jack Hathaway, MDA, JD, CCEP	Mayers Memorial Hospital (<i>Director of Quality</i>)
Amy Herlihy, RHIT, CCS	CommonSpirit Health (<i>System Director, Quality Management Systems</i>)
Linda Hightower, RN, ONC	Community Medical Center (<i>Quality Coordinator</i>)
Daniel Hoody, MD, MSc	Hennepin Healthcare (<i>Chief Medical Officer</i>)
Sandi Hyde, BSME, MSPS	LifePoint Health (<i>Assistant Vice President, Quality Data</i>)
Kacie Kleja, MBA, MS	HCA Healthcare (<i>Assistant Vice President for Clinical Analytics and Public Reporting</i>)
John Krueger, MD, MPH	Chickasaw Nation Health Systems (<i>Chief Quality Officer</i>)
Jennifer Lamprecht, MS, RN, CNL, CPHQ	Sanford Health (<i>Director of Quality Strategy</i>)
Brian Lewis, MBA	University Health System (<i>Vice President of Quality and Clinical Outcomes</i>)
Brian McCardel, MD	Sparrow Health System (<i>Surgeon, Board Quality Committee Chair</i>)
Amy Minnich, RN, MHSA	Geisinger Health System (<i>Associate Vice President of Quality and Safety</i>)
Katlyn Morin, MBA	The Carle Foundation Hospital (<i>Quality Consultant</i>)
Arby Nahapetian, MD, MPH, FACC	Adventist Health (<i>Chief Medical Officer</i>)
Henry Pitt, MD	Rutgers Cancer Institute of New Jersey (<i>Chief of Oncologic Quality</i>)
Martha Radford, MD, FACC, FAHA	NYU Langone Health (<i>Chief Quality Officer</i>)
Allen Repp, MD, MSc, SFHM, FACP	Vermont Medical Center (<i>Vice Chair for Quality</i>)
Leslie Safier, MPH	Santa Clara Valley Healthcare (<i>Director of Operational Improvements</i>)
Georgia Thomas, RN, BSN, MBA, HACP, CPHQ, FACHE	Memorial Hermann Health System (<i>System Associate Vice President of Quality</i>)
Julie Wall, RN, MBA, FACMPE	Benefis Health System (<i>Senior Vice President, Quality</i>)
Ronald Wyatt, MD, MHA	Achieving Healthcare Equity, LLC (<i>Founder, Senior Fellow IHI</i>)

Table A - 4. January 2023 PFE WG Roster

PLWG Member
Michelle Beck
Anna Schwamlein Howard
Nikki Montgomery
Linda Relyea
Patient Representative

PLWG Member
Patient Representative
Patient Representative

Table A - 5. October 2023 PFE WG Roster

PFE WG Member
Angela Buchana
Anna Schwamlein Howard
Claire Noel-Miller
David Andrews
Donte Smith
Gregory Backer
Jill Wodnick
Jodi Hansen
Latoya Hendricks
Michael Adama
Shalesse Gardner

Table A - 6. October 2023 TEP Roster

TEP Member	Title
Amy Chin, MS	Hospital for Special Surgery (<i>Hospital Administrator</i>)
Amy Minnich, RN, MHSA, CPC	Geisinger, PA (<i>Associate Vice President, Quality Safety and Patient Experience</i>)
Ashantae Okechukwu	--
Benjamin D. Pollock, PhD, MSPH	Mayo Clinic (<i>Assistant Professor of Health Services Research, Scientific Director for Data Science</i>)
Carol B. Pugh, PharmD, MS	<i>Retired Clinical Pharmacist/Biostatistician</i>
David Levine, MD, FACEP	Vizient (<i>Group Senior Vice President, Advanced Analytics & Data Science</i>)
David M. Shahian, MD	Mass. General Hospital, Society of Thoracic Surgeons (<i>Vice President, Center for Quality & Safety</i>)
Danielle Rosario	<i>HR Business Partner</i>
Erin O'Rourke, BS	America's Health Insurance Plans (<i>Executive Director, Clinical Performance and Transformation</i>)
Eugene Hsu, MD, MBA	Elevance Health (<i>Physician Executive in Clinical Quality</i>)
Itisha S. Jefferson, BA	<i>Patient/Advocate/Caregiver</i>
Jennifer Lamprecht, MS, RN, CNL, CPHQ	Sanford Health (<i>Director of Quality Strategy</i>)
John Bott, MSSW, MBA	<i>Independent Contractor</i>
John Martin, PhD, MPH	Premier, Inc. (<i>Vice President, Data Science</i>)
Jordan Russell, MPA, CPHQ	UnityPoint Health (<i>Executive Director of Quality</i>)

TEP Member	Title
Julie Wall, RN, MBA, FACMPE	Benefis Health System (<i>Senior Vice President, Quality & Patient Safety</i>)
Kacie Kleja, MBA, MS	HCA Healthcare (<i>Vice President, Clinical Data & Analytics</i>)
Lisa Freeman, BA	CT Center for Patient Safety (<i>Patient, Caregiver, Executive Director</i>)
Karl Bilimoria, MD, MS	Indiana University School of Medicine (<i>Chair, Department of Surgery, System Vice President for Quality, Executive Director of the Surgical Outcomes and Quality Improvement Center (SOQIC)</i>)
Larry Boress, MPA	National Association of Worksite Health Centers (<i>Executive Director</i>)
Marisha Burden, MD, MBA	University of Colorado School of Medicine (<i>Division Head of Hospital Medicine, Professor of Medicine</i>)
Roger Lacoy	PFCC Partners, PITCH, and PHCA Inc., (<i>Patient, Caregiver, Advocate, Board of Directors</i>)
Sandi Hyde, BSME, MSPS	Lifepoint (<i>AVP Clinical and Operations Quality Data and Regulatory Reporting</i>)
Stephanie Fishkin, PhD	Kaiser Permanente (<i>Data Analytics and Reporting Consultant</i>)

Table A - 7. November 2023 Provider WG Roster

Provider WG Member	Title
Allen Repp, MD, MSc, SFHM, FACP	Vermont Medical Center (<i>Vice Chair for Quality</i>)
Amy Aronsky, DO, MBA, FCCP, FAASM	UnitedHealthcare (<i>Senior Medical Director</i>)
Amy Herlihy, RHIT, CCS	CommonSpirit Health (<i>System Director, Quality Management Systems</i>)
Arby Nahapetian, MD, MPH, FACC	Adventist Health (<i>Chief Medical Officer</i>)
Assad Ghani MBA, MHSA, BS	University of Chicago Medicine (<i>Director of Value Based Care Models</i>)
Barbara Petersen, DNP,MSN,APRN-BC	Great Plains Health (<i>Chief Quality Officer, Administrator</i>)
Brian Lewis, MBA	University Health System (<i>Vice President, Quality and Clinical Outcomes</i>)
Brian McCardel, MD	Sparrow Health System (<i>Surgeon, Board Quality Committee Chair</i>)
Crystal DeCola, MS, RN, NE-BC	Pagosa Springs Medical Center (<i>Quality Assurance and Performance Improvement Manager</i>)
Dana Steiner, BSN, MBA, CPHQ	The Nebraska Hospital Association (<i>Nurse, Quality Director</i>)
Daniel Hoody, MD, MSc	Sanford Health of Northern Minnesota (<i>Vice President, Medical Office & Clinics Hospitalist, Department of Hospital Medicine</i>)
David M. Safley, MD	Saint Luke's Health System (<i>Vice President of Medical Affairs for Quality, Interventional Cardiologist</i>)

Provider WG Member	Title
Eric Alper, MD, FACP, SFHM, CPE	UMass Memorial Health Care (<i>Chief Quality Officer, Chief Clinical Informatics Officer</i>)
Henry Pitt, MD	Rutgers Cancer Institute of New Jersey (<i>Chief of Oncologic Quality</i>)
Jennifer Robey, CPHIMS	Mayo Clinic (<i>Manager Quality Data and Analytics</i>)
John Krueger, MD, MPH	Chickasaw Nation Health Systems (<i>Chief Quality Officer</i>)
Joy Gill, DHA, MBA, RN, CPHQ	Sheppard Pratt (<i>Clinical Director, Quality and Safety</i>)
Julie Bowen, MSN, RN, CPHQ	Gibson Area Hospital & Health Services (<i>Executive Director of Quality Assurance</i>)
Marjorie Bessel, MD	Banner Health (<i>Chief Clinical Officer</i>)
Mark Fontana, PhD, MA, M.Eng.	Hospital for Special Surgery (<i>Senior Director of Data Science</i>)
Maureen Dailey, PhD, RN, WOCN, FAAN	Daileysolution (<i>Consultant of Healthcare Quality</i>)
Ronald Wyatt, MD, MHA	Achieving Healthcare Equity, LLC (<i>Founder, Senior Fellow IHI</i>)
Suparna Dutta, MD, MS, MPH, FACP, FHM	Hartford Hospital (<i>Chair, Department of Medicine</i>)
William Jack Hathaway, MPA, JD, CCEP	Mayers Memorial Hospital (<i>Director of Quality</i>)

Table A - 8. November 2024 PFE WG Roster

PFE WG Member
Angela Buchana
Anna Schwamlein Howard
Charisse “Nikki” Montgomery
Claire Noel-Miller
David Andrews
Donte Smith
Gregory Backer
Jill Wodnick
Linda Relyea
Latoya Hendricks
Michael Adama
Shalesse Gardner

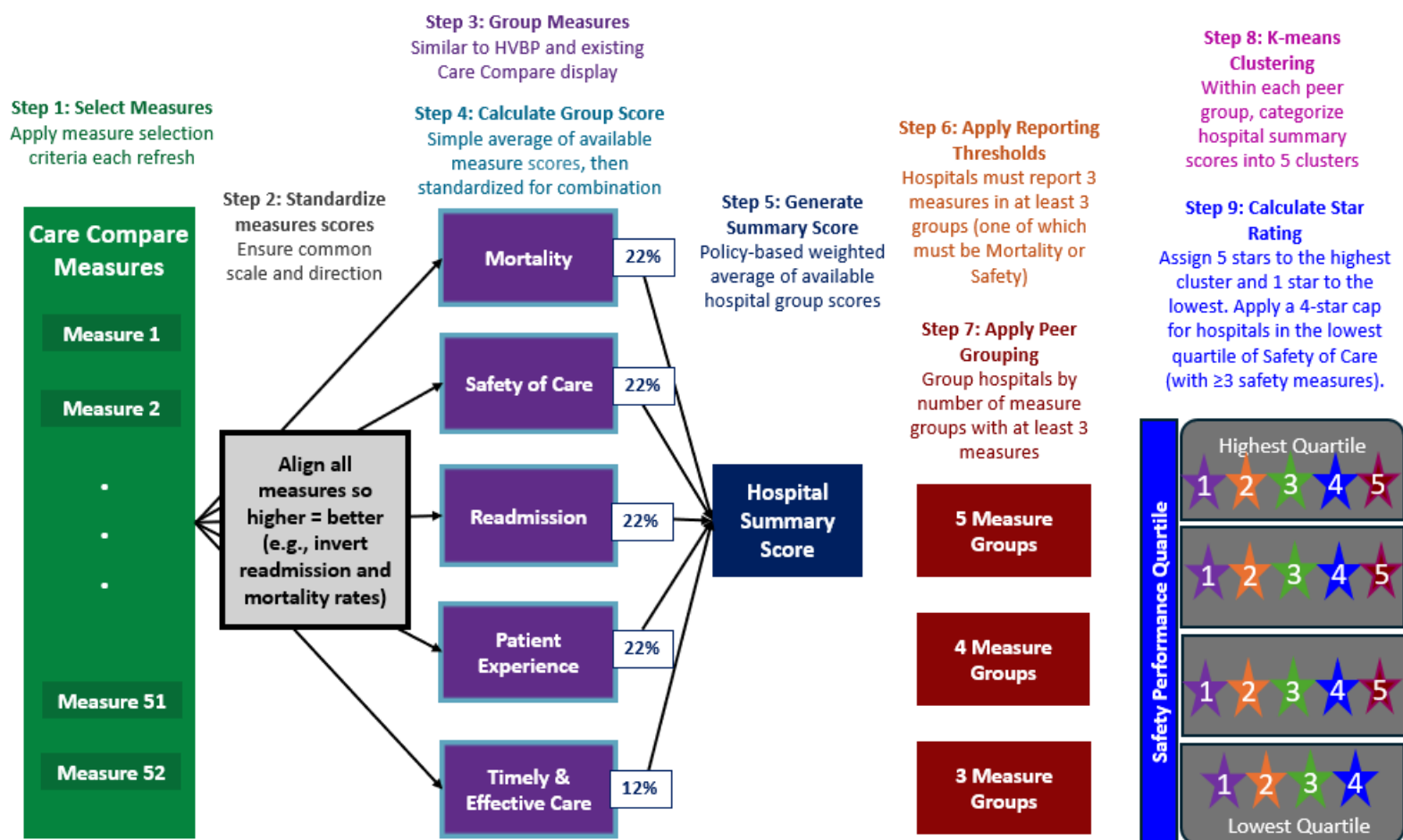
Table A - 9. December 2024 Provider WG Roster

Provider WG Member	Title
Allen Repp, MD, MSc, SFHM, FACP	Vermont Medical Center (<i>Vice Chair for Quality</i>)

Provider WG Member	Title
Amy Aronsky, DO, MBA, FCCP, FAASM	UnitedHealthcare (<i>Senior Medical Director</i>)
Amy Herlihy, RHIT, CCS	CommonSpirit Health (<i>System Director, Quality Management Systems</i>)
Arby Nahapetian, MD, MPH, FACC	Adventist Health (<i>Chief Clinical Officer</i>)
Assad Ghani MBA, MHSA, BS	University of Chicago Medicine (<i>Director of Value Based Care Models</i>)
Barbara Petersen, DNP,MSN,APRN-BC	Great Plains Health (<i>Chief Quality Officer, Administrator</i>)
Brian Lewis, MBA	University Health System (<i>Vice President, Quality and Clinical Outcomes</i>)
Brian McCardel, MD	Sparrow Health System (<i>Surgeon, Board Quality Committee Chair</i>)
Crystal DeCola, MS, RN, NE-BC	Pagosa Springs Medical Center (<i>Quality Assurance and Performance Improvement Manager</i>)
Dana Steiner, BSN, MBA, CPHQ	The Nebraska Hospital Association (<i>Nurse, Senior Director, Medicaid Quality Improvement</i>)
Daniel Hoody, MD, MSc	Sanford Health of Northern Minnesota (<i>Vice President, Medical Office & Clinics Hospitalist, Department of Hospital Medicine</i>)
David M. Safley, MD	Saint Luke's Health System (<i>Vice President of Medical Affairs for Quality, Interventional Cardiologist</i>)
Eric Alper, MD, FACP, SFHM, CPE	UMass Memorial Health Care (<i>Chief Quality Officer, Chief Clinical Informatics Officer</i>)
Henry Pitt, MD	Rutgers Cancer Institute of New Jersey (<i>Chief of Oncologic Quality</i>)
Jennifer Robey, CPHIMS	Mayo Clinic (<i>Manager Quality Data and Analytics</i>)
John Krueger, MD, MPH	Chickasaw Nation Health Systems (<i>Chief Quality Officer</i>)
Joy Gill, DHA, MBA, RN, CPHQ	Sheppard Pratt (<i>Clinical Director, Quality and Safety</i>)
Julie Bowen, MSN, RN, CPHQ	Gibson Area Hospital & Health Services (<i>Executive Director of Quality Assurance</i>)
Marjorie Bessel, MD	Banner Health (<i>Chief Clinical Officer</i>)
Mark Fontana, PhD, MA, M.Eng.	Hospital for Special Surgery (<i>Senior Director of Data Science</i>)
Maureen Dailey, PhD, RN, WOCN, FAAN	Daileysolution (<i>Consultant of Healthcare Quality</i>)
Ronald Wyatt, MD, MHA	Achieving Healthcare Equity, LLC (<i>Founder, Senior Fellow IHI</i>)
Suparna Dutta, MD, MS, MPH, FACP, FHM	Hartford Hospital (<i>Chief of Medicine</i>)
William Jack Hathaway, MPA, JD, CCEP	Mayers Memorial Hospital (<i>Director of Quality</i>)

Appendix B: Flowchart of Nine-Step Overall Star Rating Methodology

Figure B - 1. The Nine Steps of the Overall Star Rating Methodology



Appendix C: Measures Excluded from April 2026 Overall Star Rating (N=12) by Exclusion Criterion

Structural measures without evidence of an association with changes in clinical practice or improved outcomes

1. EDV-1: Emergency department volume
2. SM-7: Maternal Morbidity
3. Birthing-Friendly Hospital Designation

Non-directional measures

1. MSPB-1/SPP-1: Spending per hospital patient with Medicare
2. OP-39: Breast cancer screening recall rates
3. OP-18c: Median time psychiatric or other mental health patients spent in the emergency department before leaving from the visit
4. OP-18d: Median Time from ED Arrival to ED Departure for Discharged ED Patients – Transfer Patients

Measures publicly reported through Care Compare on Medicare.gov that hospitals are not required to submit

1. OP-31 Cataracts: Percentage of patients who had cataract surgery and had improvement in visual function within 90 days following the surgery

Overlapping measures already captured by another measure

1. READM-30-AMI: 30-day rate of readmission for heart attack patients
2. READM-30-HF: 30-day rate of readmission for heart failure patients
3. READM-30-PN: 30-day rate of readmission for pneumonia patients
4. OP-18a: Average (median) time all patients spend in the emergency department before leaving from the visit

Measures with no more than 100 hospitals reporting performance publicly

- Not applicable: No eligible measures had 100 hospitals or fewer reporting performance

Appendix D: Measures Included in the Overall Star Ratings by Measure Group (April 2026)

Mortality

1. MORT-30-AMI: 30-day death rate for heart attack patients
2. MORT-30-CABG: Death rate for coronary artery bypass graft surgery patients
3. MORT-30-COPD: Death rate for chronic obstructive pulmonary disease (COPD) patients
4. MORT-30-HF: 30-day death rate for heart failure patients
5. Hybrid HWM (MORT-30-HOSPWIDE): Death rate for patients (hospital-wide)
6. MORT-30-PN: 30-day death rate for pneumonia patients
7. MORT-30-STK: Death rate for stroke patients
8. PSI-4-SURG-COMP: Death rate among surgical inpatients with serious treatable complications

Safety of Care

1. COMP-HIP-KNEE: Rate of complications for hip and knee replacement patients
2. HAI-1: Central-line associated bloodstream infection (CLABSI)
3. HAI-2: Catheter-associated urinary tract infection (CAUTI)
4. HAI-3: Surgical site infection from colon surgery (SSI: Colon)
5. HAI-4: Surgical site infection from abdominal hysterectomy (SSI-abdominal hysterectomy)
6. HAI-5: Methicillin-resistant staphylococcus aureus (or MRSA) blood infections (Antibiotic-resistant blood infections)
7. HAI-6: Clostridium difficile (or C.diff.) infections (Intestinal infections)
8. PSI-90: Patient Safety and Adverse Events Composite

Readmission

1. EDAC-30-AMI: Acute myocardial infarction excess days in acute care (EDAC)
2. EDAC-30-HF: Heart failure excess days in acute care (EDAC)
3. EDAC-30-PN: Pneumonia excess days in acute care (EDAC)
4. READM-30-CABG: Rate of unplanned readmission after coronary artery bypass graft (CABG) surgery
5. READM-30-COPD: Rate of unplanned readmission for chronic obstructive pulmonary disease patient
6. READM-30-Hip-Knee: 30-day rate of readmission for hip and knee replacement patients
7. Hybrid HWR: Rate of readmission after discharge from hospital (hospital-wide)
8. OP-32: Facility 7-day risk standardized hospital visit rate after outpatient colonoscopy
9. OP-35 ADM: Admissions visits for patients receiving outpatient chemotherapy
10. OP-35 ED: Emergency department (ED) visits for patients receiving outpatient chemotherapy
11. OP-36: Hospital visits after hospital outpatient surgery

Patient Experience

1. H-COMP-1: Communication with nurses
2. H-COMP-2: Communication with doctors
3. H-COMP-3: Responsiveness of hospital staff
4. H-COMP-5: Communication about medicines

5. H-COMP-6: Discharge information
6. H-COMP-7: Care transition
7. H-CLEAN-HSP: Patients who reported that their room and bathroom were clean
8. H-QUIET-HSP: Patients who reported that the area around their room was quiet at night
9. H-HSP-RATING: Patients who gave their hospital a rating on a scale from 0 (lowest) to 10 (highest)
10. H-RECMND: Patients who would recommend the hospital to their friends and family
11. O-COMP-1: Patients that reported if staff gave care in a professional way and if the facility clean
12. O-COMP-2: Patients who reported if they received communication about what to expect during and after the procedure
13. O-COMP-3: Patients who reported if they were prepared for discharge and recovery
14. O-PATIENT-RATE: Patients who gave the facility a rating
15. O-PATIENT-REC: Patients who reported if they would recommend the facility

Timely and Effective Care

1. IMM-3: Percent of healthcare workers vaccinated against Influenza
2. OP-22: Percentage of patients who left the emergency department before being seen
3. OP-23: Percentage of patients who came to the emergency department with stroke symptoms who received brain scan results within 45 minutes of arrival
4. OP-29: Appropriate follow-up interval for normal colonoscopy in average risk patients
5. SEP-1: Percentage of patients who received appropriate care for severe sepsis and septic shock
6. OP-18b: Average time patients spent in the emergency department before being sent home
7. OP-8: Outpatients with low back pain who had an MRI without trying recommended treatments first, such as physical therapy
8. OP-10: Outpatient CT scans of the abdomen that were “combination” (double) scans
9. OP-13: Medicare patients who got cardiac imaging stress tests to screen for surgical risk before low-risk outpatient surgery
10. SAFE_USE_OF_OPIOIDS: Percentage of patients who were prescribed 2 or more opioids or an opioid and benzodiazepine concurrently at discharge

Appendix E: List of All Measures and Their Contribution to Hospitals' Overall Star Rating (April 2021)

To address questions regarding the impact of single measures on hospitals' overall scores, we here reproduce findings from report v4.1 for information purposes. Similar analyses over multiple other years (results not reported here) demonstrated broadly consistent tendencies.

Contribution of Single Measures to Measure Group Scores

The use of a simple average approach to measure score combination is sensitive to heterogeneity in measure reporting between hospitals. While most hospitals report most measures in any given measure group, select hospitals report fewer measure groups and fewer measures, which in turn results in a relatively higher contribution, or effective weight, for a given measure towards measure group scores and hospital summary scores.

Of the 3,355 hospitals with an Overall Star Rating, 320 reported on a single measure in at least one measure group. Of these hospitals, the very rare circumstance in which a hospital reported a single measure in two measure groups only occurred for 10 hospitals. The most common measure group to include a single measure was the Safety of Care measure group. 275 hospitals that reported on a single measure in the Safety of Care measure group were assigned an Overall Star Rating. Among these 275 hospitals, the three Safety of Care measures that were most commonly included as the lone measure in the Safety of Care measure group were Patient Safety Indication 90 (PSI-90), Hip/Knee Complications of Care, and Catheter-Associated Urinary Tract Infection ([Table E - 1](#)).

When examining the contribution of each individual measure towards the hospital summary score of all 3,355 hospitals' Overall Star Rating, we found that the contribution, or effective weight, of any single measure score to hospitals' Overall Star Rating was below 5% for half of all included hospitals. For 75% of hospitals, no individual measure accounted for more than 7% of their Overall Star Rating.

Closer examination of each measures' contribution to hospitals' Overall Star Rating revealed that performance on Overall Star Ratings was not driven by any one specific measure. In general, the distribution of single measure weight was right-skewed. For 75% of hospitals, no specific measure accounted for more than 5.5% of their Overall Star Rating. At the tail end of this distribution, for 95% of hospitals one specific measure did not account for more than 11% of hospitals' Overall Star Rating. Thus, while any one measure contributed no more than 22% of the Overall Star Rating for 95% of hospitals, one specific measure, like Patient Safety Indication 90, was only responsible for 11% of the Overall Star Rating for 95% of hospitals.

Table E - 1. Number of Hospitals Reporting a Single Measure in a Measure Group

Measure group	# Hospitals with One Measure in a Measure Group
Safety of Care	275
PSI 90	105
Hip/Knee Complications	83

Measure group	# Hospitals with One Measure in a Measure Group
Clostridium Difficile	81
Catheter-Associated Urinary Tract Infection	6
Readmission	20
Hospital-Wide 30-Day	20
Timely and Effective Care	18
OP-10 Abdomen Computed Tomography Use of Contrast Material	6
IMM-3 Health Care Personnel Influenza Vaccination	6
ED-2b Admit Decision Time to Emergency Department Departure Time for Admitted Patients	3
OP-13 Cardiac Imaging for Preoperative Risk Assessment for Non-Cardiac Low-Risk Surgery	3
Mortality	17
Pneumonia 30-Day	13
PSI 4	3
Stroke 30-Day	1
Patient Experience	0
Total*	320

* 10 hospitals had one measure in two measure groups

Table E - 2. List of All Measures and Their Contribution to Hospitals' Overall Star Rating (April 2021)

Measure Group	Measure	Measure Contribution to Overall Hospital Star Rating for All Hospitals						
		Min	5 th Pctl	25 th Pctl	50 th Pctl	75 th Pctl	95 th Pctl	Max
Mortality	Overall Group Score **	3.1%	3.1%	3.1%	4.4%	5.5%	11.0%	28.2%
	MORT_30_PN ^{††}	0.0%	3.1%	3.1%	3.7%	5.5%	9.8%	28.2%
	MORT_30_STK	0.0%	0.0%	0.0%	3.1%	3.7%	5.5%	22.0%
	PSI_4_SURG_COMP	0.0%	0.0%	0.0%	3.1%	3.1%	3.7%	22.0%
	MORT_30_COPD	0.0%	3.1%	3.1%	3.7%	5.5%	9.4%	14.1%
	MORT_30_HF	0.0%	3.1%	3.1%	3.7%	5.5%	9.4%	14.1%
	MORT-30-AMI	0.0%	0.0%	0.0%	3.1%	3.7%	4.4%	11.0%
	MORT-30-CABG	0.0%	0.0%	0.0%	0.0%	3.1%	3.1%	7.3%
Safety of Care	Overall Group Score	2.8%	2.8%	3.1%	3.7%	7.3%	22.0%	28.2%
	PSI_90_SAFETY	0.0%	0.0%	2.8%	3.1%	5.5%	11.0%	28.2%
	HAI_6	0.0%	0.0%	2.8%	3.1%	5.5%	11.0%	28.2%
	COMP_HIP_KNEE	0.0%	0.0%	0.0%	3.1%	4.4%	11.0%	28.2%
	HAI_2	0.0%	0.0%	0.0%	2.8%	3.1%	5.5%	28.2%

** Each measure group's contribution to the Hospitals' Overall rating is based off its weighted average. See Step 4: Weighted Average of Groups to Calculate Summary Scores

†† Each measure's contribution to the Hospitals' Overall rating is based on the combination of the simple average of eligible measures within the group score, and the group score's weighted average. See Step 3: Calculation of Group Scores Using a Simple Average of Measure Scores

Measure Group	Measure	Measure Contribution to Overall Hospital Star Rating for All Hospitals						
		Min	5 th Pctl	25 th Pctl	50 th Pctl	75 th Pctl	95 th Pctl	Max
	HAI_3	0.0%	0.0%	0.0%	2.8%	3.1%	4.0%	13.1%
	HAI_1	0.0%	0.0%	0.0%	2.8%	3.1%	4.4%	9.8%
	HAI_5	0.0%	0.0%	0.0%	2.8%	3.1%	3.7%	9.8%
	HAI_4	0.0%	0.0%	0.0%	0.0%	0.0%	2.8%	9.4%
Readmission	Overall Group Score	2.0%	2.0%	2.2%	2.8%	3.1%	7.1%	28.2%
	READM_30_HOSP_WIDE	0.0%	2.0%	2.2%	2.8%	3.1%	7.1%	28.2%
	EDAC_30_HF	0.0%	2.0%	2.2%	2.4%	3.1%	6.5%	14.1%
	OP_36	0.0%	0.0%	2.0%	2.2%	2.8%	3.7%	14.1%
	READM_30_HIP_KNEE	0.0%	0.0%	0.0%	2.2%	2.8%	3.1%	14.1%
	EDAC_30_PN	0.0%	2.0%	2.2%	2.8%	3.1%	6.5%	13.1%
	OP_32	0.0%	0.0%	2.0%	2.4%	3.1%	5.6%	13.1%
	READM_30_COPD	0.0%	2.0%	2.2%	2.4%	3.1%	5.6%	13.1%
	OP_35_ADM	0.0%	0.0%	0.0%	0.0%	2.2%	2.4%	7.9%
	OP_35_ED	0.0%	0.0%	0.0%	0.0%	2.2%	2.4%	7.9%
	EDAC_30_AMI	0.0%	0.0%	0.0%	2.0%	2.4%	2.8%	7.3%
	READM_30_CABG	0.0%	0.0%	0.0%	0.0%	2.0%	2.4%	4.4%
Patient Experience	Overall Group Score	2.8%	2.8%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_1	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_2	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_3	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_5	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_6	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_COMP_7	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_RATING+H_RECMMND/2	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
	H_CLEAN+H_QUIET/2	0.0%	0.0%	2.8%	2.8%	2.8%	2.8%	4.9%
Timely and Effective Care	Overall Group Score	0.9%	0.9%	1.1%	1.2%	1.5%	4.0%	15.4%
	ED_2b	0.0%	0.9%	1.0%	1.2%	1.5%	3.6%	15.4%
	IMM_3	0.0%	0.0%	0.0%	1.0%	1.2%	3.0%	15.4%
	OP_13	0.0%	0.0%	0.0%	1.0%	1.1%	1.5%	15.4%
	OP_10	0.0%	0.9%	1.0%	1.2%	1.5%	3.6%	12.0%
	OP_18b	0.0%	0.9%	1.0%	1.2%	1.5%	3.1%	7.7%
	OP_22	0.0%	0.0%	1.0%	1.2%	1.3%	2.6%	7.7%
	SEP_1	0.0%	0.0%	1.0%	1.1%	1.3%	2.0%	7.7%
	PC_01	0.0%	0.0%	0.0%	1.0%	1.2%	1.5%	7.7%
	OP_8	0.0%	0.0%	0.0%	0.0%	1.0%	1.2%	7.7%
	OP_23	0.0%	0.0%	0.0%	0.0%	1.1%	1.3%	7.1%
	OP_33	0.0%	0.0%	0.0%	0.0%	0.0%	1.1%	6.0%
	OP_30	0.0%	0.0%	0.9%	1.1%	1.2%	1.7%	5.1%
	OP_29	0.0%	0.0%	0.9%	1.1%	1.2%	1.7%	4.3%
	OP_3b	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	4.3%

Appendix F: Reliability and Validity Analysis (April 2021)

Validity Testing

To address questions regarding the impact of single measures on hospitals' overall scores, we here reproduce findings from report v4.1 for information purposes. Similar analyses over multiple other years (results not reported here) demonstrated broadly consistent tendencies.

Pairwise Comparison of Scores between Overall Hospital Star Ratings

To test the validity of the Overall Star Rating, generated using *k*-means clustering to complete convergence, CMS conducted an analysis that describes the distribution of group scores for hospitals in each Overall Star Rating category using October 2020 data. CMS tested the association between the mean group score for each Overall Star Rating category and the mean group score for each of the other Overall Star Rating categories for each measure group, using Tukey's method for multiple comparisons.¹⁰ This analysis demonstrated statistically different measure group scores between each Overall Star Rating category in all groups, supporting the ability of *k*-means clustering to distinguish hospital performance across the five clusters (See [Tables F1-F5: Pairwise Comparison of Mean Group Scores between Star Classes](#)).

CMS found statistically significant differences in all pair-wise comparisons of measure group scores between each Overall Star Rating category for the 5-measure group peer group. Similarly, there were statistically significant differences for most pair-wise comparisons for the 4-measure peer group (41/50, including all comparisons in Readmission and all but one in both Safety of Care and Patient Experience) and for the 3-measure peer group (38/50, including all comparisons in the Mortality and Readmission groups). CMS did not observe a situation in which a higher measure group score was significantly associated with a lower Overall Star Rating.

Test for the Linear Trend of Measure Group Scores

To further confirm the validity of the Overall Star Rating methodology, CMS examined the relationship across the mean measure groups scores for each star rating category, comparing the slope across the mean group scores for each category to zero. CMS found a statistically significant ($p < 0.001$) linear trend in measure group scores for all groups. The results of this analysis, using October 2020 data, are depicted as box plots by measure group ([Figure F.1](#)).

Reliability Testing

The CMS Overall Star Rating is a summary measure of hospital performance which is affected by both the individual hospital's performance and also relative performance of peers. As a part of reliability testing of the Overall Star Rating, CMS examined the effect of excluding a subset of reporting hospitals on the Overall Star Rating. This analysis assesses the extent of sensitivity of the Overall Star Rating to hospitals either not meeting reporting thresholds or voluntary suppression of star ratings by hospitals. To assess the sensitivity of the Overall Star Rating to the inclusion or exclusion of hospitals, CMS performed a simulation where 10 percent of hospitals were randomly selected for exclusion from the dataset and Overall Star Ratings were recalculated; this was repeated 1000 times with the Overall Star Ratings recalculated for each iteration.

The mean (minimum-maximum) difference in Overall Star Rating for individual hospitals after randomly removing 10 percent of hospitals from the sample and running 1000 iterations of star ratings was 0.03 (0-1 stars). The Overall Star Rating was identical in 96.8 percent of iterations. This suggests that individual hospital's star ratings are dependent largely on individual performance and only minimally affected by random perturbations to the hospital quality scores reported in the entire dataset.

Summary of Testing

The analyses conducted by CMS support the methodological enhancements in v4.0, which were driven by a desire to simplify the Overall Star Rating methodology. This simplification enhanced the Overall Star Rating interpretability and uptake while retaining the validity and reliability of results sought by CMS for public reporting.

In nearly all instances, hospitals with greater numbers of stars performed significantly better on the measures included in the Overall Star Rating than hospitals with fewer stars. This relationship existed for each peer group, indicating that the Overall Star Rating increases as hospital measure group scores increase. Additionally, the Overall Star Rating is robust to the exclusion of reporting 10 percent fewer hospitals, indicating that hospital performance on the measures drives their Overall Star Rating.

As the distribution of hospital performance evolves and/or further updates are made to the Overall Star Rating methodology, CMS will continue to test the methodology and potential improvements to the Overall Star Rating.

Pairwise Comparison of Mean Group Scores between Star Classes

Tables F.1-F5 presents the results of CMS's pairwise comparisons of mean group scores between the Star Rating classes using October 2020 data. Differences marked with an asterisk (*) are statistically significant (that is, $p < 0.05$).

Table F - 1. Pairwise Comparison: Mean group score difference by Overall Star Rating and Peer Group

Star Comparison	3-group peer group (n=308)	4-group peer group (n=536)	5-group peer group (n=2,465)
5-4	0.57*	0.38	0.58*
5-3	1.02*	0.95*	0.97*
5-2	1.61*	1.27*	1.25*
5-1	2.18*	1.25*	1.59*
4-3	0.45*	0.56*	0.39*
4-2	1.04*	0.89*	0.67*
4-1	1.61*	0.87*	1.01*
3-2	0.59*	0.32*	0.28*
3-1	1.16*	0.3	0.62*
2-1	0.57*	-0.02	0.34*

Table F - 2. Pairwise Comparison of Star Classes using Safety of Care Mean Group Scores

Star Comparison	3-group peer group (n=200)	4-group peer group (n=500)	5-group peer group (n=2,465)
5-4	0.71	0.58	0.29*
5-3	1.23*	1.11*	0.52*
5-2	1.25*	1.79*	0.85*
5-1	2.23*	3.86*	1.38*
4-3	0.53	0.54*	0.23*
4-2	0.55	1.21*	0.56*
4-1	1.53*	3.29*	1.09*
3-2	0.02	0.67*	0.33*
3-1	1.00*	2.75*	0.86*
2-1	0.98*	2.08*	0.53*

Table F - 3. Pairwise Comparison of Star Classes using Readmission Mean Group Scores

Star Comparison	3-group peer group (n=336)	4-group peer group (n=553)	5-group peer group (n=2,465)
5-4	0.58*	0.89*	0.49*
5-3	1.08*	1.21*	0.94*
5-2	1.60*	1.62*	1.41*
5-1	2.66*	2.45*	2.29*
4-3	0.50*	0.32*	0.46*
4-2	1.02*	0.72*	0.92*
4-1	2.08*	1.56*	1.80*
3-2	0.52*	0.41*	0.46*
3-1	1.58*	1.24*	1.34*
2-1	1.06*	0.83*	0.88*

Table F - 4. Pairwise Comparison of Star Classes using Patient Experience Mean Group Scores

Star Comparison	3-group peer group (n=68)	4-group peer group (n=543)	5-group peer group (n=2,465)
5-4	0.97*	0.38	0.46*
5-3	1.26*	0.81*	0.89*
5-2	1.62*	1.46*	1.41*
5-1	2.84*	2.01*	2.01*
4-3	0.29	0.44*	0.43*
4-2	0.65	1.08*	0.95*
4-1	1.87*	1.63*	1.55*
3-2	0.36	0.65*	0.52*
3-1	1.58*	1.19*	1.12*
2-1	1.22	0.55*	0.60*

Table F - 5. Pairwise Comparison of Star Classes using Timely & Effective Care Mean Group Scores

Star Comparison	3-group peer group (n=331)	4-group peer group (n=552)	5-group peer group (n=2,465)
5-4	0.35	0.01	0.22*
5-3	0.56*	0.25	0.39*
5-2	1.03*	0.44*	0.59*
5-1	1.03*	1.74*	0.98*
4-3	0.21	0.25	0.17*
4-2	0.69*	0.44*	0.37*
4-1	0.68*	1.73*	0.76*
3-2	0.47*	0.19	0.20*
3-1	0.47	1.48*	0.59*
2-1	-0.01	1.29*	0.39*

Linear Trend of Overall Star Rating Measure Group Scores across Star Categories (April 2021)

Figure F - 1. Mortality Group Scores Across Star Categories

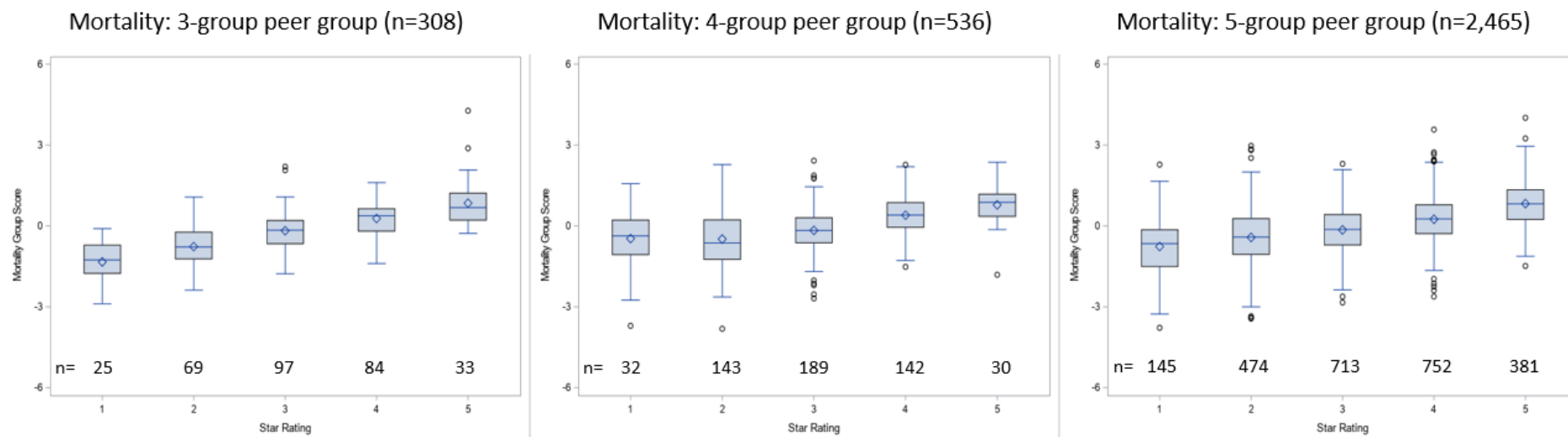


Figure F - 2. Safety of Care Group Scores Across Star Categories

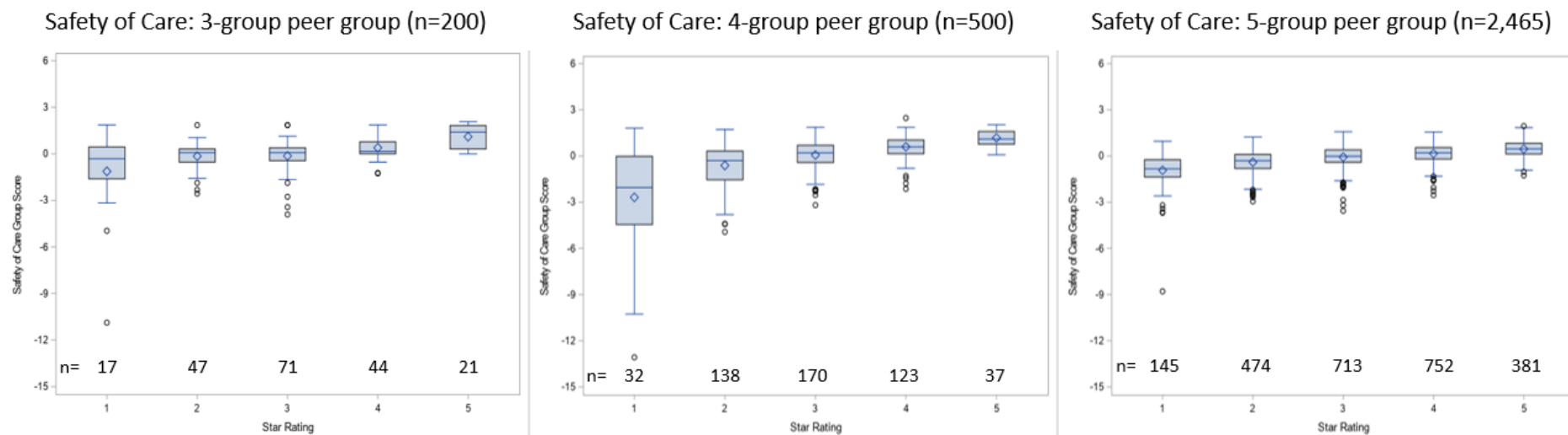


Figure F - 3. Readmission Group Scores Across Star Categories

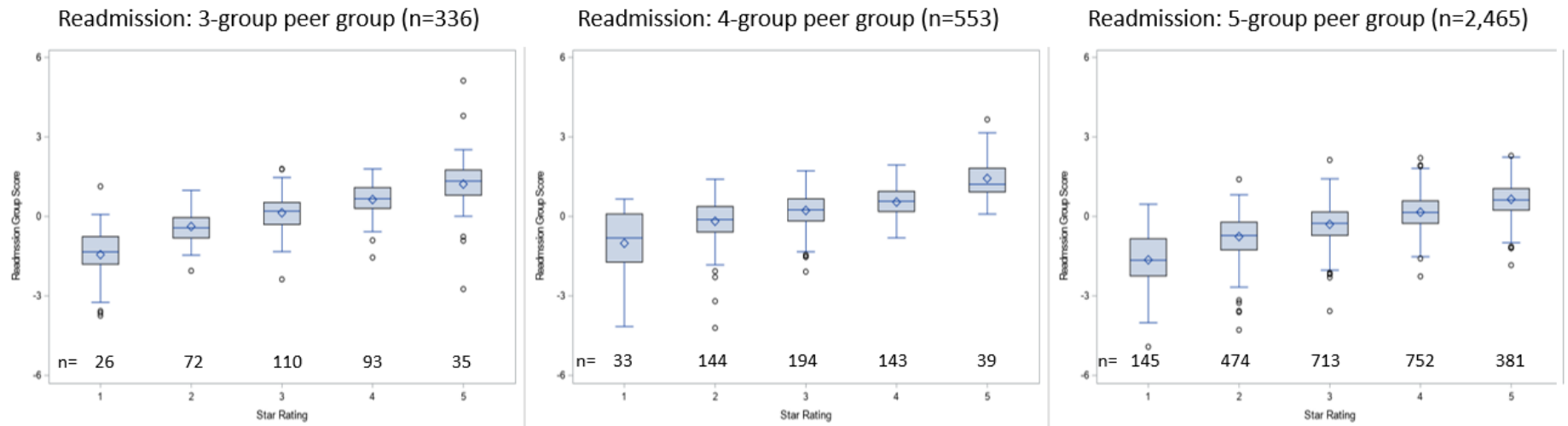


Figure F - 4. Patient Experience Group Scores Across Star Categories

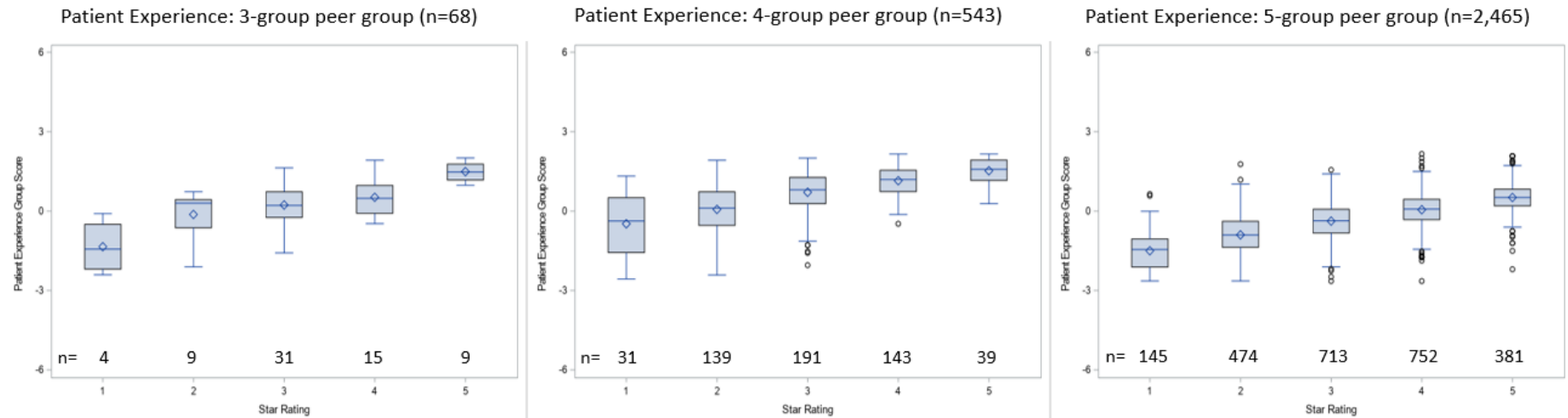


Figure F - 5. Timely & Effective Care Group Scores Across Star Categories

